Eliciting causal models – Analysis

```
#options(device = svglite::svglite)

library(svglite)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(dagitty)
library(tidygraph)
```

```
Attaching package: 'tidygraph'

The following object is masked from 'package:dagitty':

    convert

The following object is masked from 'package:stats':

    filter
```

```
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

```
library(patchwork)
library(ggsci)
library(ggokabeito)
# library(ggpubfigs)
#library(cowplot)
```

Data import

Sessions

Graphical Causal Models

Modelling sessions were originally run on Loopy, then transcribed into DAGitty (online) to generate the following `dot` language syntax for importing into `R`. A better way of doing this is under development. Code hidden for this pdf render, can be inspected in the .qmd file.

Tidy graphs

Data imported as DAGitty objects, but need to be in tidygraph format for many calculations and visualisations.

```
# helper to get back door paths straight away - not working for all graphs yet
path_to_edges <- function(path) {
  tokens <- str_split(path, " ")[[1]]
  nodes  <- tokens[c(TRUE, FALSE)]
  arrows <- tokens[c(FALSE, TRUE)]
```

```
  tibble(
    from = ifelse(arrows == "->", nodes[-length(nodes)], nodes[-1]),
    to   = ifelse(arrows == "->", nodes[-1], nodes[-length(nodes)])
  )
}

open_backdoor_edgelist <- function(dag) {
  as_tibble(paths(backDoorGraph(dag), limit = 1e4)) |>
        filter(open) |>
      pull(paths) |>
      map_df(path_to_edges) |>
      distinct()
}
```

```
dag_to_tidy_graph <- function(dag){
    # Extract edges
    edges <- as.data.frame( dagitty::edges(dag) )
    edges$edge_w = 1
    nodes <- tibble(name = unique(c(edges$v, edges$w)))

    # Build tidygraph
    tg <- tbl_graph(nodes = nodes, edges = edges, directed = TRUE)
    return(tg)
}
```

Creating a list of all the DAGs in tidy format:

```
tdag_s1 <- dag_to_tidy_graph(dag_s1)
tdag_s2 <- dag_to_tidy_graph(dag_s2)
tdag_s3 <- dag_to_tidy_graph(dag_s3)
tdag_s4 <- dag_to_tidy_graph(dag_s4)
tdag_s5 <- dag_to_tidy_graph(dag_s5)
tdag_s6 <- dag_to_tidy_graph(dag_s6)
tdag_s7 <- dag_to_tidy_graph(dag_s7)
tdag_s8 <- dag_to_tidy_graph(dag_s8)

quick_list_of_all_nodes <- unique(c(tdag_s1 |> pull(name),
                                    tdag_s2 |> pull(name),
                                    tdag_s3 |> pull(name),
                                    tdag_s4 |> pull(name),
                                    tdag_s5 |> pull(name),
```

```
                                    tdag_s6 |> pull(name),
                                    tdag_s7 |> pull(name),
                                    tdag_s8 |> pull(name)
                                    )
                                  )
```

Coding

Node Classifiers

Coding the nodes into rough groups. There are two levels:

- `node_group` is the first aggregation into slightly coarser groups of nodes.
- `node_type` is an even more coarse grouping into a handful of types.

Also some nicer names of the fields: - `node_nice_name` is a longer more explanatory name for plotting and putting in tables. - `node_group_nice_name` a longer more explanatory name for node groups, as above.

```
name_node_nice_name <- tribble(
    ~name, ~nice_name,
    "CD", "Any system data",
    "CD.Att", "Attendance",
    "CD.LMS", "LMS activity",
    "CD.LMS.code", "Code artifacts",
    "CD.LMS.write", "Writing artifacts",
    "CD.Str", "Course Data",
    "Grade", "Assessment grade",
    "Grade.Coms", "Mark - Communication",
    "Grade.Exam", "Mark - Exam",
    "Grade.Interp", "Mark - Interpretation",
    "Grade.Proc", "Mark - Learning process",
    "Grade.Quiz", "Mark - Quiz",
    "Grade.Ref", "Mark - Learning reflection",
    "Grade.Tech", "Mark - Technical",
    "H", "Help (general)",
    "H.AI", "Help from AI (any)",
    "H.AI.Agent", "Help from AI (agent)",
    "H.AI.evil", "Misuse of AI",
    "H.AI.evil.code", "Misuse of AI (code)",
    "H.AI.evil.write", "Misuse from AI (write)",
```

```
    "H.AI.Fb", "Help from AI (feedback)",
    "H.AI.good", "Help from AI (to learn)",
    "H.AI.Res", "Help from AI (research)",
    "H.AI.Study", "Help from AI (revision)",
    "H.Ment", "Help from mentor",
    "H.other.evil", "Cheating - non-AI",
    "H.Peer", "Help from peers",
    "H.Tut", "Help from teacher / tutor",
    "H.Tut.Fb", "Feedback from teacher / tutor",
    "LD", "Learning design",
    "LD.AIguide", "Guidance on AI use",
    "LD.Task", "Task design",
    "LD.Task.Code", "Coding task design",
    "LD.Task.Grp", "Groupwork task design",
    "LD.Task.Other", "Other task design",
    "LD.Task.Prac", "Practical task design",
    "LD.Task.Write", "Writing task design",
    "S.Act", "Student decisions / actions",
    "S.Act.AIstrat", "Student AI strategy",
    "S.Act.Fb", "Student acts on feedback",
    "S.Act.Fb.Post", "Student acts on feedback between subjects",
    "S.Act.Frm", "Student does formative tasks",
    "S.Act.Perf", "Student performance on task",
    "S.Act.Time", "Student commits time",
    "S.Aff.Integ", "Student acting with integrity",
    "S.Aff.Joy", "Student enjoyment",
    "S.Aff.MindSet", "Student mindset",
    "S.Aff.Safe", "Student feels safe",
    "S.Aff.WrkLd", "Student workload pressure",
    "S.Att.Eq", "Student equity group",
    "S.Att.HSk", "Student seeks help",
    "S.Att.Nro", "Student neuro-divergent",
    "S.Att.SocCap", "Student social capital",
    "S.Cp.Aca", "Student academic skill",
    "S.Cp.EvJdg", "Student evaluative judgement",
    "S.Cp.Fb", "Student feedback literacy",
    "S.Cp.Grp", "Student groupwork capability",
    "S.Cp.Ind", "Student individual work capability",
    "S.Cp.Lrn", "Student metacognative learning capabilities",
    "S.Cp.SRL", "Student SRL",
    "S.Kn", "Student subject knowledge",
    "S.Kn.Coms", "Student communications knowledge",
```

```
    "S.Kn.Interp", "Student interpretation knowledge",
    "S.Kn.Prior", "Student prior knowledge",
    "S.Kn.Tech", "Student technical knowledge",
    "T.Act.UseAI", "Teacher use of AI",
    "T.Aff.Integ", "Teacher acts with integrity",
    "T.Aff.WrkLd", "Teacher workload pressure",
    "T.Cp.Inst", "Teacher instruction quality",
    "T.Cp.KnStu", "Teacher knows students",
    "T.Cp.Mark", "Teacher reliably marks",
    "T.Kn.AI", "Teacher AI literacy",
    "T.Kn.Ass", "Teacher assessment literacy",
    "T.Kn.Sub", "Teacher subject knowledge"
)

node_group_nice_name <- tribble(
    ~node_group, ~nice_name,
    "Assessments", "Preliminary assessments",
    "CD.Att", "Student attendance",
    "CD.LMS", "LMS activity",
    "CD.Str", "Course Data",
    "Grade.Outcome", "Assessment grade",
    "H.AI.evil", "Misuse of AI",
    "H.AI.good", "Help from AI",
    "H.Cheat", "Cheating - non-AI",
    "H.Peer", "Help from peers",
    "H.Tut", "Help from teacher / tutor",
    "LD", "Learning design",
    "S.Act", "Student decisions / actions",
    "S.Att", "Student attributes",
    "S.Cp", "Student metacognitive skill",
    "S.EM", "Student emotion / motivation",
    "S.Kn", "Student knowledge",
    "S.Kn.Prior", "Student prior knowledge",
    "T.Act", "Teacher use of AI",
    "T.Cp", "Teacher capabilities",
    "T.EM", "Teacher emotion / motivation",
    "T.Kn", "Teacher knowledge"
)
```

Coding node groups

```
code_node_group <- function(v) {
    case_when(
        # Course data - Str, LMS --------------
        v == "CD" ~ "CD.Str", # was blending LMS and Str, but original graph was more about s
        str_detect(v, "CD\\.LMS") ~ "CD.LMS",
        v == "CD.Att" ~ "CD.Att",
        v == "CD.Str" ~ "CD.Str",
        # Grade data --------------------------
        v == "Grade" ~ "Grade.Outcome",
        str_detect(v, "Grade\\.") ~ "Assessments",
        # Kinds of help -----------------------
        # - - - Help from teacher or tutor
        v == "H" ~ "H.Tut", # closest match for the only broad category one
        v == "H.Tut" ~ "H.Tut",
        v == "H.Ment" ~ "H.Tut",
        v == "H.Tut.Fb" ~ "H.Tut",
        # - - - Help from peers
        v == "H.Peer" ~ "H.Peer",
        # - - - Help from AI (good or bad usage)
        str_detect(v, "H\\.AI\\.evil") ~ "H.AI.evil",
        str_detect(v, "H\\.AI") ~ "H.AI.good",
        # - - - Other nefarious help (i.e. cheating)
        v == "H.other.evil" ~ "H.Cheat",
        # Learning Design --------------------
        str_detect(v, "^LD") ~ "LD",
        # Student activity -------------------
        str_detect(v, "^S\\.Act") ~ "S.Act",
        # Student attributes (generally unchangable) ---
        str_detect(v, "^S\\.Att") ~ "S.Att",
        # Student metacognitive capability (metacognition)
        str_detect(v, "^S\\.Cp") ~ "S.Cp",
        # Student emotion / motivation
        str_detect(v, "^S\\.Aff") ~ "S.EM",
        # Student knowledge (cognition)
        v == "S.Kn.Prior" ~ "S.Kn.Prior", # Prior knowledge treated as it's own node
        str_detect(v, "^S\\.Kn") ~ "S.Kn",
        # Teacher knowledge (cognition)
        str_detect(v, "^T\\.Kn") ~ "T.Kn",
        # Teacher capabilities (metacognition)
        str_detect(v, "^T\\.Cp") ~ "T.Cp",
```

```
        # Teacher emotion / motivation
        str_detect(v, "^T\\.Aff") ~ "T.EM",
        # Teacher actions / decisions
        str_detect(v, "^T\\.Act") ~ "T.Act",
        TRUE ~ v

    )
}
```

Coding node types

```
code_node_type <- function(n) {
    fl <- str_extract(n, "^.{1}")
    case_when(
        fl == "S" ~ "Student",
        fl == "T" ~ "Teacher",
        fl == "C" ~ "Learning data",
        fl == "H" ~ "Help",
        fl == "L" ~ "Learning design",
        fl == "G" ~ "Assessment data"
    ) |>
        factor(levels = c("Assessment data", "Learning data", "Learning design", "Teacher",
}
```

Coding node measurability

From the node groups, describe how easy (or not) they are to measure. This is in three levels:

- Directly: Data is available or easy enough to get directly for the node
- Proxy: Node is not directly measurable, but with work a direct proxy for the node could be found.
- Intangible: Node is not currently measurable, and unlikely to have a direct proxy.

By "direct proxy" I mean one step away in the causal graph. So if X, Y is not measurably, but Z is then $X \to Z$ would mean that X is measurable by proxy. However if at best we can only justify $X \to Y \to Z$ then X is intangible.

```
code_node_measurability <- function(node) {
    case_when(
        # Student
```

```
        node == "S.Att.SocCap" ~ "PO", # check
        node == "S.Att.Nro" ~ "O", # check
        node == "S.Att.Eq" ~ "DW", # check
        node == "S.Att.HSk" ~ "L", # check
        node == "S.Act.AIstrat" ~ "L",
        node == "S.Act.Fb" ~ "L",
        node == "S.Act.Fb.Post" ~ "L",
        str_detect(node, "^S\\.Act") ~ "PO",
        str_detect(node, "^S") ~ "L", # Latent
        # Help
        node == "H.other.evil" ~ "L",
        node == "H.Peer" ~ "L",
        node == "H.AI.evil.code" ~ "PO", # Potentially observable
        node == "H.AI.evil.write" ~ "PO",
        node == "H.AI.evil.write" ~ "PO",
        str_detect(node, "^H\\.AI") ~ "PO",
        str_detect(node, "^H") ~ "L",
        # Assessment
        node == "Grade" ~ "DW",
        node == "Grade.Exam" ~ "DW",
        str_detect(node, "^Grade") ~ "O",
        # Learning data
        node == "CD.LMS" ~ "DW",
        str_detect(node, "^CD\\.LMS") ~ "O",
        node == "CD.Att" ~ "PO",
        node == "CD.Str" ~ "DW", # check
        node == "CD" ~ "PO", # check
        # Learning design
        str_detect(node, "^LD") ~ "PO",
        # Teacher
        node == "T.Act.UseAI" ~ "PO",
        node == "T.Aff.WrkLd" ~ "PO", # check this, not act with integ as L
        str_detect(node, "^T") ~ "L",
        node == "Assessments" ~ "DW"
    ) |>
        factor(levels = c("DW", "O", "PO", "L"))
}
```

Graph processing

Graph helper functions

Convention is that these all start with `g_` and accept a tidygraph object as input.

Graph manipulator functions

```
# exposures(dag_s1)
library(igraph)
```

```
Attaching package: 'igraph'

The following object is masked from 'package:tidygraph':

    groups

The following object is masked from 'package:dagitty':

    edges

The following objects are masked from 'package:lubridate':

    %--%, union

The following objects are masked from 'package:dplyr':

    as_data_frame, groups, union

The following objects are masked from 'package:purrr':

    compose, simplify

The following object is masked from 'package:tidyr':

    crossing
```

```
The following object is masked from 'package:tibble':

    as_data_frame

The following objects are masked from 'package:stats':

    decompose, spectrum

The following object is masked from 'package:base':

    union
```

```r
library(dagitty)

g_edgelist <- function(g){
    # tidygraph to tibble of edges
    g %E>%
        as_tibble() %>%
        mutate(
            from = g %N>% pull(name) %>% .[from],
            to   = g %N>% pull(name) %>% .[to]
        ) %>%
        select(from, to)
}

g_merge_nodes <- function(g, regex_from, string_to, no.self.loops = TRUE) {
    # merges nodes in a graph by renaming edgelist
    g_el <- g_edgelist(g)
    g_el_m <- g_el |>
        mutate(from = str_replace(from, regex_from, string_to),
               to = str_replace(to, regex_from, string_to)) |>
        distinct()
    tbl_graph(edges = g_el_m)
}

# same but for dags
dag_merge_nodes <- function(dag, mapping) {
    # mapping is a named list, like: c("Old.var" = "Merged.var", "Old.var2" = "Merged.var")
    g <- graph_from_data_frame(dagitty::edges(dag))
    old_names <- V(g)$name
    # Replace names using mapping if present, else keep original
    new_names <- ifelse(old_names %in% names(mapping),
```

```
                        mapping[old_names],
                        old_names)
    V(g)$name <- unname(new_names)
    new_g <- simplify(g, remove.loops = TRUE, remove.multiple = TRUE)

    # Convert back to dagitty
    edges_m <- igraph::as_data_frame(new_g, what="edges")
    dag_str <- paste0("dag { ", paste0(edges_m$from, " -> ", edges_m$to, collapse="; "), " }
    g_m <- dagitty(dag_str)
    g_m
}

g_get_adjustment_sets <- function(g, exposure = "S.Kn", outcome = "Grade") {
    if (!is_acyclic(as.igraph(g))) {
        return(list("Cyclic" = "No adjustment sets"))
    }
  # Extract edges
  edges <- g_edgelist(g)

  # Build dagitty syntax
  edges_str <- edges %>%
    mutate(rel = paste0(from, " -> ", to)) %>%
    pull(rel)

  dagitty_str <- paste("dag {", paste(edges_str, collapse = "\n  "), "}", sep="\n")
  dagitty_obj <- dagitty(dagitty_str)

  # Compute adjustment sets
  sets <- adjustmentSets(dagitty_obj, exposure = exposure, outcome = outcome)
  as.list(sets)
}
```

Graph metrics functions

```
# would like pagerank with "Grade" excluded

g_node_metrics <- function(g) {
    pagerank_no_outcome <-
        g %N>%
        filter(name != "Grade") |>
```

```
        mutate(pagerank_no_outcome = centrality_pagerank(damping = 0.85)) |>
        as_tibble() |>
        select(name, pagerank_no_outcome)

    g %N>%
        mutate(
            node_w = 1,
            node_group = code_node_group(name),
            node_type = code_node_type(name),
            measurability = code_node_measurability(name),
            degree = centrality_degree(),
            btw = centrality_betweenness(),
            btw_rank = rank(-btw, ties.method = "min"),   # rank (highest = 1)
            btw_rel_rank = (btw_rank - 1) / (n() - 1),
            btw_rel = 1 - btw_rel_rank,
            eigen = centrality_eigen(),
            pagerank = centrality_pagerank(damping = 0.5) # lowers the influence of sinks
        ) |>
        as_tibble() |>
        left_join(pagerank_no_outcome, by = "name") |>
        mutate(pagerank_no_outcome = replace_na(pagerank_no_outcome, 0))
}
```

```
g_summary_metrics <- function(g){
    pagerank_no_outcome <-
        g %N>%
        filter(name != "Grade") |>
        mutate(pagerank_no_outcome = centrality_pagerank(damping = 0.5)) |>
        as_tibble() |>
        select(name, pagerank_no_outcome)

    n.df <- g %N>%
        mutate(
            pgrnk = centrality_pagerank(damping = 0.5),
            ng = code_node_group(name),
            m = code_node_measurability(name)) |>
        as_tibble() |>
        left_join(pagerank_no_outcome, by = "name")

    # p_traced <- mean(n.df$m == "Traced")
    # p_untraced <- mean(n.df$m == "Untraced")
    # pgrnk_traced <- sum(n.df |> filter(m == "Traced") |> pull(pgrnk))
```

```
    # pgrnk_untraced <- sum(n.df |> filter(m == "Untraced") |> pull(pgrnk))
    # pgrnk_no_outcome_traced <- sum(n.df |> filter(m == "Traced") |> pull(pagerank_no_outcor
    # pgrnk_no_outcome_untraced <- sum(n.df |> filter(m == "Untraced") |> pull(pagerank_no_ou

    p_dw <- mean(n.df$m == "DW")
    p_o <- mean(n.df$m == "O")
    p_po <- mean(n.df$m == "PO")
    p_l <- mean(n.df$m == "L")
    measure_metric <- (3 * p_dw + 2 * p_o + 1 * p_po ) / 3

    graph_metrics <- list(
        order     = gorder(g), # number of nodes
        size      = gsize(g), # number of edges
        density   = edge_density(g),
        diameter  = diameter(g), # longest shortest path
        mean_path = mean_distance(g),
        clustering= transitivity(g, type = "global"),
        acyclic = igraph::is_acyclic(as.igraph(g)),
        measurability = measure_metric,
        p_dw = p_dw,
        p_o = p_o,
        p_po = p_po,
        p_l = p_l
    )
    graph_metrics
}
```

Generating a range of edge metrics - comparisons across models. Want to label edges in the unified DAG as 'contested' if they switch direction between some models.

```
tdags <- list(tdag_s1, tdag_s2, tdag_s3, tdag_s4,
              tdag_s5, tdag_s6, tdag_s7, tdag_s8)

g_labelled_edgelist <- function(g) {
    m <- as.character(substitute(g)) |>
        str_remove("^tdag_")
    g_edgelist(g) |>
        mutate(Model = m) |>
        select(Model, everything())
}

g_labelled_nodelist <- function(g) {
```

```
    m <- as.character(substitute(g)) |>
        str_remove("^tdag_")
     g %N>%
         as_tibble() |>
         select(name) |>
        mutate(Model = m) |>
        select(Model, everything())
}

g_possible_edges <- function(g) {
    m <- as.character(substitute(g)) |>
        str_remove("^tdag_")
    g_labelled_nodelist(g) |>
        rename(from = name) |>
        mutate(Model = m) |>
        cross_join(g_labelled_nodelist(g) |>
                       select(-Model) |>
                       rename(to = name))
}

all_possible_edges <- bind_rows(
    g_possible_edges(tdag_s1),
    g_possible_edges(tdag_s2)
) |>
    bind_rows(g_possible_edges(tdag_s3)) |>
    bind_rows(g_possible_edges(tdag_s4)) |>
    bind_rows(g_possible_edges(tdag_s5)) |>
    bind_rows(g_possible_edges(tdag_s6)) |>
    bind_rows(g_possible_edges(tdag_s7)) |>
    bind_rows(g_possible_edges(tdag_s8))

max_possible_edges <- all_possible_edges |>
    filter(from != to) |>
    count(from, to) |>
    rename(max_n = n)

all_nodes <- bind_rows(g_labelled_nodelist(tdag_s1),
                       g_labelled_nodelist(tdag_s2)) |>
    bind_rows(g_labelled_nodelist(tdag_s3)) |>
    bind_rows(g_labelled_nodelist(tdag_s4)) |>
    bind_rows(g_labelled_nodelist(tdag_s5)) |>
    bind_rows(g_labelled_nodelist(tdag_s6)) |>
```

```
    bind_rows(g_labelled_nodelist(tdag_s7)) |>
    bind_rows(g_labelled_nodelist(tdag_s8))

all_edges <- bind_rows(g_labelled_edgelist(tdag_s1),
                       g_labelled_edgelist(tdag_s2)) |>
    bind_rows(g_labelled_edgelist(tdag_s3)) |>
    bind_rows(g_labelled_edgelist(tdag_s4)) |>
    bind_rows(g_labelled_edgelist(tdag_s5)) |>
    bind_rows(g_labelled_edgelist(tdag_s6)) |>
    bind_rows(g_labelled_edgelist(tdag_s7)) |>
    bind_rows(g_labelled_edgelist(tdag_s8))
reverses <- tibble(Model = all_edges$Model, from = all_edges$to, to = all_edges$from)

g_all_descendants_edge_list <- function(g, m) {
    # Get descendant edges
    expanded <- distances(g) < Inf
    closure_edges <- which(expanded, arr.ind = TRUE)

    expanded_edges <- data.frame(
        Model = m,
        from = V(g)[closure_edges[,1]]$name,
        to   = V(g)[closure_edges[,2]]$name
    ) |> dplyr::filter(from != to)
    return(expanded_edges)
}

all_descendants <-
    bind_rows(
        g_all_descendants_edge_list(tdag_s1, "s1"),
        g_all_descendants_edge_list(tdag_s2, "s2"),
    ) |>
    bind_rows(g_all_descendants_edge_list(tdag_s3, "s3")) |>
    bind_rows(g_all_descendants_edge_list(tdag_s4, "s4")) |>
    bind_rows(g_all_descendants_edge_list(tdag_s5, "s5")) |>
    bind_rows(g_all_descendants_edge_list(tdag_s6, "s6")) |>
    bind_rows(g_all_descendants_edge_list(tdag_s7, "s7")) |>
    bind_rows(g_all_descendants_edge_list(tdag_s8, "s8")) |>
    as_tibble()


all_edges_summary <-
    all_possible_edges |>
```

```
    filter(from != to) |>
    left_join(all_edges |> mutate(in_graph = TRUE)) |>
    mutate(in_graph = replace_na(in_graph, FALSE)) |>
    left_join(all_descendants |> mutate(descendant = TRUE)) |>
    mutate(descendant = replace_na(descendant, FALSE))
```

```
Joining with `by = join_by(Model, from, to)`
Joining with `by = join_by(Model, from, to)`
```

```
all_edges_sum <-
    all_edges_summary |>
    group_by(from, to) |>
    summarise(n = sum(in_graph),
              n_desc = sum(descendant)) |>
    arrange(desc(n)) |>
    ungroup()
```

```
`summarise()` has grouped output by 'from'. You can override using the
`.groups` argument.
```

```
all_edges_with_metrics <-
    all_edges_sum |>
    filter(n > 0) |>
    inner_join(max_possible_edges) |>
    mutate(
        edge_p_w = n / max_n,
        edge_dp_w = n_desc / max_n) |>
    mutate(edge_w = n)
```

```
Joining with `by = join_by(from, to)`
```

```
# edge_not_chosen <-
#     all_possible_edges |>
#     inner_join(edges_with_metrics |>
#                    filter(edge_p_w < 1) , relationship = "many-to-many") |>
#     anti_join(all_edges) |>
#     select(Model, from, to) |>
#     distinct() |>
#     left_join(
#         all_descendants |>
```

```
#             # anti_join(all_edges) |>
#             mutate(is_descendant = TRUE)) |>
#     mutate(is_descendant = replace_na(is_descendant, FALSE))

# edges_with_metrics <-
#     edges_with_metrics |>
#     left_join(edge_not_chosen |>
#                   filter(is_descendant) |>
#                   group_by(from, to) |>
#                   summarise(n_mediated = sum(is_descendant))) |>
#     mutate(n_mediated = replace_na(n_mediated, 0)) |>
#     distinct()

switches <- inner_join(all_edges, reverses, by = c("from", "to"),
                       relationship = "many-to-many") |>
    arrange(Model.x)

edges_with_metrics <-
    all_edges_with_metrics |>
    left_join(switches |>
                  select(from, to) |> mutate(switched = "contested")) |>
    mutate(switched = replace_na(switched, "aligned"))
```

```
Joining with `by = join_by(from, to)`
```

```
edges_with_metrics |>
    count(switched) |>
    mutate(n = if_else(switched == "aligned", n, n/2)) |> # contested are counted twice due t
    mutate(p = n / sum(n))
```

```
# A tibble: 2 x 3
  switched      n      p
  <chr>     <dbl>  <dbl>
1 aligned     181 0.968
2 contested     6 0.0321
```

```
edges_with_metrics |>
    filter(switched == "contested" | n > 1) |>
    count(switched) |>
    mutate(n = if_else(switched == "aligned", n, n/2)) |> # contested are counted twice due t
    mutate(p = n / sum(n))
```

```
# A tibble: 2 x 3
  switched      n     p
  <chr>     <dbl> <dbl>
1 aligned       7 0.538
2 contested     6 0.462
```

Also examining counts of the kinds of paths:

```
count_paths <- function(dag) {
  tibble(
    total     = nrow(as_tibble(paths(dag, limit = 1e4))),
    backdoor  = nrow(as_tibble(paths(backDoorGraph(dag), limit = 1e4))),
    frontdoor = nrow(as_tibble(paths(dag, directed = TRUE, limit = 1e4)))
  )
}
```

Creating merged graph

```
library(ggraph)

merged_edges <- g_edgelist(tdag_s1) |>
    bind_rows(g_edgelist(tdag_s2)) |>
    bind_rows(g_edgelist(tdag_s3)) |>
    bind_rows(g_edgelist(tdag_s4)) |>
    bind_rows(g_edgelist(tdag_s5)) |>
    bind_rows(g_edgelist(tdag_s6)) |>
    bind_rows(g_edgelist(tdag_s7)) |>
    bind_rows(g_edgelist(tdag_s8)) |>
    count(from, to) |> ungroup() |> rename(edge_w = n) |>
    inner_join(edges_with_metrics)
```

```
Joining with `by = join_by(from, to, edge_w)`
```

```
merged_nodes <- tdag_s1 %N>% as_tibble() |>
    bind_rows(tdag_s2 %N>% as_tibble()) |>
    bind_rows(tdag_s3 %N>% as_tibble()) |>
    bind_rows(tdag_s4 %N>% as_tibble()) |>
    bind_rows(tdag_s5 %N>% as_tibble()) |>
    bind_rows(tdag_s6 %N>% as_tibble()) |>
```

```
    bind_rows(tdag_s7 %N>% as_tibble()) |>
    bind_rows(tdag_s8 %N>% as_tibble()) |>
    count(name) |> ungroup() |> rename(node_w = n) |>
    mutate(node_group = code_node_group(name)) |>
    mutate(node_type = code_node_type(name)) |>
    mutate(measurable = code_node_measurability(name)) |>
    left_join(name_node_nice_name, by = "name")

# cognitive, metacognitive, emotion / motivation, collaboration
merged_all_graph <- tbl_graph(nodes = merged_nodes, edges = merged_edges)
# View(merged_nodes |> arrange(name))
```

Coarsen graphs

This is based on the coding of the nodes into node groups.

```
g_coarsen_dag_by_ng <- function(g){
    # Creates a "coarser" graph by merging all nodes in the same node group
    old_nodes <- g %N>% as_tibble()

    if (!("node_w" %in% names(old_nodes))) {
        old_nodes <- old_nodes |>
            mutate(node_w = 1)
    }

    new_nodes <- old_nodes |>
        mutate(measurability = code_node_measurability(name)) |>
        mutate(measurability_metric = case_when(
            measurability == "DW" ~ 1,
            measurability == "O" ~ 0.7,
            measurability == "PO" ~ 0.4,
            measurability == "L" ~ 0
        )) |>
        mutate(weighted_measurability = measurability_metric * node_w) |>
        mutate(name = code_node_group(name),
               node_type = code_node_type(name)) |>
        group_by(name, node_type) |>
        summarise(
            node_w = sum(node_w),
            measurability_metric = sum(weighted_measurability) / sum(node_w),
            .groups = "drop"
```

```
        )

    new_edges <- g |>
        g_edgelist() |>
    mutate(across(from:to, code_node_group)) |>
    count(from, to) |>
    rename(edge_w = n)

    tbl_graph(nodes = new_nodes, edges = new_edges)
}
```

A more brutal coarsening by node type.

```
g_coarsen_dag_by_type <- function(g){
    new_nodes <- g %N>%
        as_tibble() |>
        mutate(name = code_node_type(name)) |>
        count(name) |>
        rename(node_w = n)

    new_edges <- g |>
        g_edgelist() |>
    mutate(across(from:to, code_node_type)) |>
    count(from, to) |>
    rename(edge_w = n)

    tbl_graph(nodes = new_nodes, edges = new_edges)
}
```

Here we merge nodes into node groups before generating plot later. This has the potential to likely introduce loops depending on how the constructs were built.

```
merged_coarse_models <-
    tbl_graph(
        nodes = all_nodes |>
            mutate(name = code_node_group(name)) |>
            distinct(Model, name) |>
            count(name) |>
            rename(node_w = n) |>
            inner_join(node_group_nice_name |> rename(name = node_group)),
        edges = all_edges |>
```

```
            mutate(across(from:to, code_node_group)) |>
            distinct() |>
            count(from, to) |>
            rename(edge_w = n)
    )
```

```
Joining with `by = join_by(name)`
```

Graph metrics

Measurability, complexity

Model level metrics

```
all_paths_summary <- tibble(
  Model = str_c("S", 1:8),
  DAG   = list(dag_s1, dag_s2, dag_s3, dag_s4, dag_s5, dag_s6, dag_s7, dag_s8)
) %>%
  mutate(
    counts = map(DAG, ~ count_paths(.x))
  ) %>%
  unnest(counts)

all_paths <- tibble(
    Model = str_c("S", 1:8),
    Paths.df = list(
        as_tibble(paths(dag_s1,  limit = 5000)),
        as_tibble(paths(dag_s2,  limit = 5000)),
        as_tibble(paths(dag_s3,  limit = 5000)),
        as_tibble(paths(dag_s4,  limit = 5000)),
        as_tibble(paths(dag_s5,  limit = 5000)),
        as_tibble(paths(dag_s6,  limit = 5000)),
        as_tibble(paths(dag_s7,  limit = 5000)),
        as_tibble(paths(dag_s8,  limit = 5000))
    )) |>
    mutate(
        n_paths = map_int(Paths.df, nrow),
        n_open = map_int(Paths.df, ~sum(pull(.,open))),
        p_open = map_dbl(Paths.df, ~mean(pull(.,open)))
```

```
    )

# Yes, this really should be implemented using map_df or something more elegant
dag_complexity_metrics <-
    bind_cols(tibble(Model = "S1"),
          g_summary_metrics(tdag_s1) |>
              as_tibble()) |>
    bind_rows(
        bind_cols(tibble(Model = "S2"),
          g_summary_metrics(tdag_s2) |>
              as_tibble())
    ) |>
    bind_rows(
        bind_cols(tibble(Model = "S3"),
          g_summary_metrics(tdag_s3) |>
              as_tibble())
    ) |>
    bind_rows(
        bind_cols(tibble(Model = "S4"),
          g_summary_metrics(tdag_s4) |>
              as_tibble())
    ) |>
    bind_rows(
        bind_cols(tibble(Model = "S5"),
          g_summary_metrics(tdag_s5) |>
              as_tibble())
    ) |>
    bind_rows(
        bind_cols(tibble(Model = "S6"),
          g_summary_metrics(tdag_s6) |>
              as_tibble())
    ) |>
    bind_rows(
        bind_cols(tibble(Model = "S7"),
          g_summary_metrics(tdag_s7) |>
              as_tibble())
    ) |>
    bind_rows(
        bind_cols(tibble(Model = "S8"),
          g_summary_metrics(tdag_s8) |>
              as_tibble())
    ) |>
```

```
    inner_join(
        tibble(
            Model = str_c("S", 1:8),
            n_adj_sets = map_int(list(dag_s1, dag_s2, dag_s3, dag_s4,
                                      dag_s5, dag_s6, dag_s7, dag_s8),
                                 ~length(adjustmentSets(.)))
        ), by = "Model"
    ) |>
    inner_join(all_paths_summary, by = "Model")
```

Graph dissimilarity

Taken from suggestions in Grames et al. (2022) this calculates the graph dissimilarity from Shieber et al. (2017).

```
source("d-measures.R") # Functions from Grames et al.
```

Creating distance matrices for both the fine and coarse graphs. The coarse seem to give a better 'feel' for the differences in the models. In the end a mixture distnace is created using the average of the dissimilarity at each level: *node*, *group* and *type*.

```
coarsened_tdags <- map(tdags, g_coarsen_dag_by_ng)
coarsest_tdags <- map(tdags, g_coarsen_dag_by_type)

d_matrix_fine <-
    data.frame(
        S1 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s1)),
        S2 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s2)),
        S3 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s3)),
        S4 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s4)),
        S5 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s5)),
        S6 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s6)),
        S7 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s7)),
        S8 = map_dbl(tdags, ~calc_dissimilarity(., tdag_s8))
        )
```

```
Warning: `shortest.paths()` was deprecated in igraph 2.0.0.
i Please use `distances()` instead.
```

```r
d_matrix_coarse <-
    data.frame(
        S1 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[1]])),
        S2 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[2]])),
        S3 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[3]])),
        S4 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[4]])),
        S5 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[5]])),
        S6 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[6]])),
        S7 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[7]])),
        S8 = map_dbl(coarsened_tdags, ~calc_dissimilarity(., coarsened_tdags[[8]]))
        )

d_matrix_coarsest <-
    data.frame(
        S1 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[1]])),
        S2 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[2]])),
        S3 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[3]])),
        S4 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[4]])),
        S5 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[5]])),
        S6 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[6]])),
        S7 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[7]])),
        S8 = map_dbl(coarsest_tdags, ~calc_dissimilarity(., coarsest_tdags[[8]]))
        )

d_matrix <- # trying coarse only - seems a better representation
    tibble(from = str_c("S", 1:8)) |>
    bind_cols(d_matrix_coarse)

d_matrix <- bind_cols( # weighted average: 3 parts coarse, 1 part fine. Fine alone seems to r
    tibble(from = str_c("S", 1:8)),
    (d_matrix_fine + d_matrix_coarse + d_matrix_coarsest) / 3)


d_edges <- d_matrix |>
    pivot_longer(cols = -from,
                 names_to = "to") |>
  filter(value > 0) |>
  rename(edge_d = value) |>
    mutate(from = factor(from), to = factor(to)) |>
    filter(as.integer(from) < as.integer(to)) |>
    arrange(edge_d) |>
    mutate(edge_w = 1 / edge_d)
```

Node group level metrics

```
all_nodes_with_metrics <- g_node_metrics(tdag_s1) |>
    mutate(Model = "S1") |>
    bind_rows(
        g_node_metrics(tdag_s2) |>
            mutate(Model = "S2")
    ) |>
    bind_rows(
        g_node_metrics(tdag_s3) |>
            mutate(Model = "S3")
    ) |>
    bind_rows(
        g_node_metrics(tdag_s4) |>
            mutate(Model = "S4")
    ) |>
    bind_rows(
        g_node_metrics(tdag_s5) |>
            mutate(Model = "S5")
    ) |>
    bind_rows(
        g_node_metrics(tdag_s6) |>
            mutate(Model = "S6")
    ) |>
    bind_rows(
        g_node_metrics(tdag_s7) |>
            mutate(Model = "S7")
    ) |>
    bind_rows(
        g_node_metrics(tdag_s8) |>
            mutate(Model = "S8")
    )
```

Idea: can we get each nodes path length to a measurable (downstream / descendant) node?

```
g_min_distance_to_node <- function(g, node) {
    g %N>%
        mutate(dist_to = distances(as.igraph(.),
                                   v = V(as.igraph(.)),
                                to = node,
                                mode = "out")[1,]) |>
```

```
        as_tibble()
}

g_distances <- function(g) {
    d <-
        distances(as.igraph(g), mode = "out")
    row_nodes <- rownames(d)
    bind_cols(tibble(from = row_nodes),
              as_tibble(d)) |>
        pivot_longer(-from, names_to = "to", values_to = "distance")
}

g_distancce_to_traced <- function(g) { #old
    g_distances(g) |>
        inner_join(all_nodes_with_metrics |>
                       # filter(measurability == "Traced") |>
                       filter(measurability == "DW") |>
                       select(to = name) |>
                       distinct()) |>
        group_by(from) |>
        summarise(dist_to_trace = min(distance)) |>
        rename(name = from)
}

all_model_nodes_distance_to_traced <-
    tibble(Model = "S1") |> bind_cols(g_distancce_to_traced(tdag_s1)) |>
    bind_rows(tibble(Model = "S2") |> bind_cols(g_distancce_to_traced(tdag_s2))) |>
    bind_rows(tibble(Model = "S3") |> bind_cols(g_distancce_to_traced(tdag_s3))) |>
    bind_rows(tibble(Model = "S4") |> bind_cols(g_distancce_to_traced(tdag_s4))) |>
    bind_rows(tibble(Model = "S5") |> bind_cols(g_distancce_to_traced(tdag_s5))) |>
    bind_rows(tibble(Model = "S6") |> bind_cols(g_distancce_to_traced(tdag_s6))) |>
    bind_rows(tibble(Model = "S7") |> bind_cols(g_distancce_to_traced(tdag_s7))) |>
    bind_rows(tibble(Model = "S8") |> bind_cols(g_distancce_to_traced(tdag_s8)))
```

```
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
Joining with `by = join_by(to)`
```

```
all_nodes_median_dist_to_traded <-
    all_model_nodes_distance_to_traced |>
    group_by(name) |>
    summarise(m_dist_to_trace = median(dist_to_trace))

all_node_groups_distance_to_traced <-
    all_model_nodes_distance_to_traced |>
    mutate(node_group = code_node_group(name)) |>
    group_by(Model, node_group) |>
    summarise(m_dist_to_trace = mean(dist_to_trace))
```

```
`summarise()` has grouped output by 'Model'. You can override using the
`.groups` argument.
```

Joining metrics and session data

```
gcm_metrics <-
    inner_join(session_data, dag_complexity_metrics, by = "Model")
```

Visualisations

Session and other

Figure: Time spent with model and modelling utility

```
# denser option - radial arc bars for observable
library(scales)
library(ggforce)
library(patchwork)
library(cowplot)
```

```
Attaching package: 'cowplot'
```

```
The following object is masked from 'package:patchwork':

    align_plots
```

```
The following object is masked from 'package:lubridate':

    stamp
```

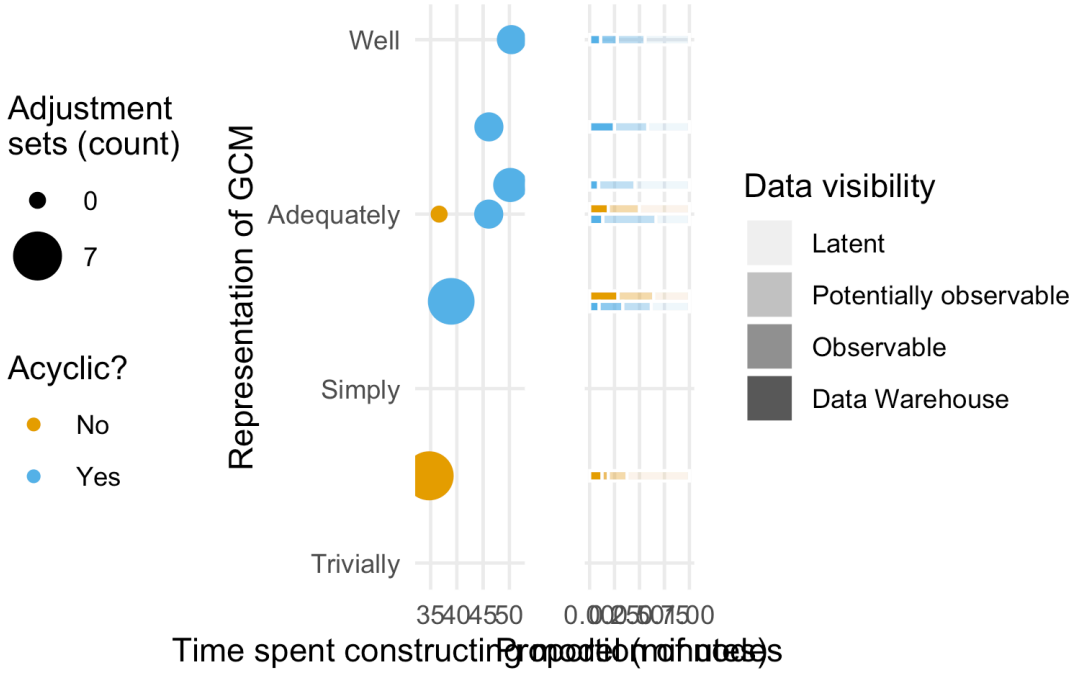
```r
hue1 <- palette.colors(3, palette = "Okabe-Ito")[[2]]
hue2 <- palette.colors(3, palette = "Okabe-Ito")[[3]]

# Main plot - hide all legends
p_validity_and_time <- gcm_metrics |>
    mutate(
        x_duration = as.numeric(Modelling.duration, units = "minutes"),
        y_validity = GCM.validity
    ) |>
    mutate(acyclic = if_else(acyclic, "Yes", "No")) |>
    ggplot(aes(
        color = acyclic,
        size = n_adj_sets
    )) +
    geom_point(aes(
        x = x_duration,
        y = y_validity)) +
    scale_y_continuous(
        limits = c(1, 4.05),
        breaks = c(1, 2, 3, 4),
        labels = c("Trivially", "Simply", "Adequately", "Well"),
        name = "Representation of GCM") +
    scale_x_continuous(
        limits = c(33, 52),  # was 55 to  Keep original limits for the data
        breaks = c(35, 40, 45,50),
        name = "Time spent constructing model (minutes)") +
    scale_size(range = c(2, 7),
               breaks = range(gcm_metrics$n_adj_sets),
               name = "Adjustment\nsets (count)") +
    scale_color_manual(values = c(hue1, hue2), name = "Acyclic?") +
    scale_fill_manual(values = c(hue1, hue2), guide = "none") +
    theme_minimal() +
    # coord_fixed(ratio = 2
    #             # , clip = "off"
    #             ) +
    theme(
        legend.position = "left",  # Hide legends from main plot
        # panel.grid.major.x = element_blank(),  # Remove vertical gridlines
```

```
        panel.grid.minor = element_blank(),
        plot.margin = margin(10, 10, 10, 10)
    )


p_visibility <- gcm_metrics |>
    mutate(
        cp_dw = p_dw,
        cp_o = p_dw + p_o,
        cp_po = p_dw + p_o + p_po,
        cp_l = p_dw + p_o + p_po + p_l # should equal 1
    ) |>
    mutate(GCM.validity = case_when(
        GCM.validity == 3 & acyclic ~ GCM.validity - 0.03,
        GCM.validity == 3 & !acyclic ~ GCM.validity + 0.03,
        GCM.validity == 2.5 & acyclic ~ GCM.validity - 0.03,
        GCM.validity == 2.5 & !acyclic ~ GCM.validity + 0.03,
        TRUE ~ GCM.validity
    )) |>
    pivot_longer(c(starts_with("cp_"), starts_with("p_")), names_to = "node_vis", values_to =
    mutate(node_visibility = case_when(
        str_detect(node_vis, "p_dw") ~ "Data Warehouse",
        str_detect(node_vis, "p_o") ~ "Observable",
        str_detect(node_vis, "p_po") ~ "Potentially observable",
        str_detect(node_vis, "p_l") ~ "Latent"
    ) |>
        ordered(levels = c(
            "Latent",
            "Potentially observable",
            "Observable",
            "Data Warehouse"
        ))) |>
    mutate(prop_type = if_else(str_detect(node_vis, "^cp_"), "cp", "p")) |>
    pivot_wider(names_from = prop_type, values_from = proportion, id_cols = c(Model,acyclic,
    mutate(acyclic = if_else(acyclic, "Yes", "No")) |>
    ggplot(aes(
        color = acyclic
    )) +
    geom_rect(aes(
        xmin = cp - p,
        xmax = cp,
        ymin = GCM.validity - 0.03,
```

```
        ymax = GCM.validity + 0.03,
        fill = acyclic,
        alpha = node_visibility),
        color = "white") +
    scale_y_continuous(
        limits = c(1, 4.05),
        breaks = c(1, 2, 3, 4),
        labels = c("Trivially", "Simply", "Adequately", "Well"),
        name = "Representation of GCM",
        position = "right") +
    scale_alpha_discrete(name = "Data visibility") +
    scale_fill_manual(values = c(hue1, hue2), guide = "none") +
    theme_minimal() +
    # coord_fixed(ratio = 0.15
    #             # , clip = "off"
    #             ) +
    xlab("Proportion of nodes") +
    theme(
        legend.position = "right",  # Hide legends from main plot
        # panel.grid.major.x = element_blank(),  # Remove vertical gridlines
        panel.grid.minor = element_blank(),
        plot.margin = margin(10, 10, 10, 10),
        axis.title.y = element_blank(),
        axis.text.y = element_blank()
    )
```

```
Warning: Using alpha for a discrete variable is not advised.
```

```
p_time_and_representation_1 <- p_validity_and_time + p_visibility
p_time_and_representation_1
```




```
ggsave("figures/FIG--time-modelling-and-representation.jpg",
       plot = p_time_and_representation_1,
       width = 10, height = 5,
       dpi = 300)

ggsave("figures/FIG--time-modelling-and-representation.svg",
       plot = p_time_and_representation_1,
       width = 25, height = 10, units = "cm")
```

This version displays all in one (is a little dense)

```
# denser option - radial arc bars for observable
library(scales)
library(ggforce)
library(patchwork)
library(cowplot)

hue1 <- palette.colors(3, palette = "Okabe-Ito")[[2]]
hue2 <- palette.colors(3, palette = "Okabe-Ito")[[3]]
```

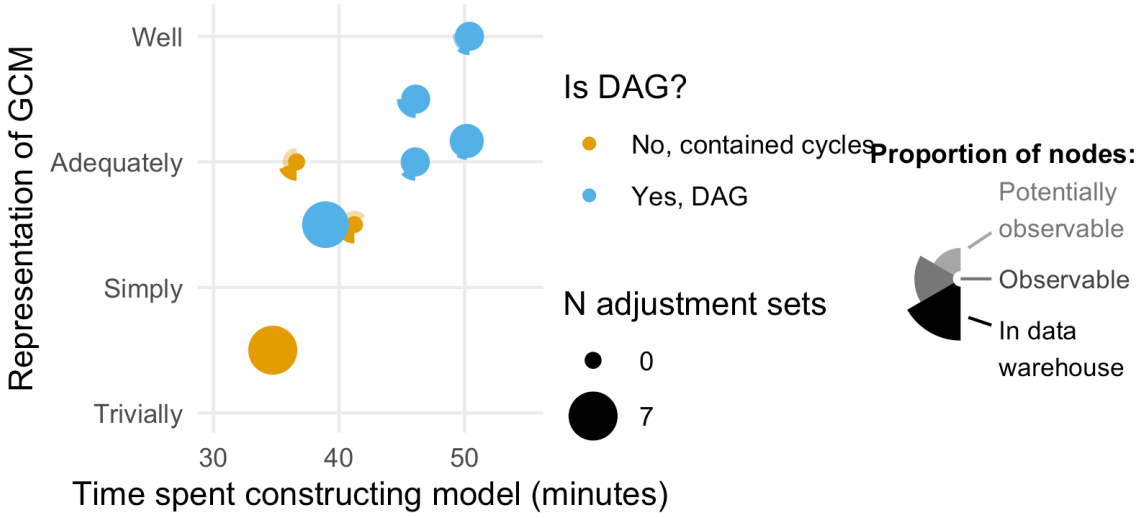
```r
# Main plot - hide all legends
p_main <- gcm_metrics |>
    mutate(
        x_duration = as.numeric(Modelling.duration, units = "minutes"),
        y_validity = GCM.validity * 10 + 20
    ) |>
    mutate(acyclic = if_else(acyclic, "Yes, DAG", "No, contained cycles")) |>
    ggplot(aes(
        color = acyclic,
        size = n_adj_sets
    )) +
    geom_arc_bar(aes(
        x0 = x_duration,
        y0 = y_validity,
        r0 = 0,
        r = 1.1,
        start = pi + 2 * (p_dw + p_o + p_po) * pi,
        end = 3 * pi
    ),
        fill = "white",
        linewidth = 0,
    alpha = 0.8, show.legend = FALSE) +
    geom_arc_bar(aes(
        x0 = x_duration,
        y0 = y_validity,
        fill = acyclic,
        size = 1,
        r0 = 0,
        r = 1.5,
        start = pi,
        end = pi + 2 * p_dw * pi
    ),
        linewidth = 0,
    alpha = 1, show.legend = FALSE) +
    geom_arc_bar(aes(
        x0 = x_duration,
        y0 = y_validity,
        fill = acyclic,
        r0 = 0,
        r = 1.3,
        start = pi + 2 * p_dw * pi,
        end = pi + 2 * (p_dw + p_o) * pi
```

```
    ),
        linewidth = 0,
    alpha = 0.6, show.legend = FALSE) +
    geom_arc_bar(aes(
        x0 = x_duration,
        y0 = y_validity,
        fill = acyclic,
        r0 = 0,
        r = 1.1,
        start = pi + 2 * (p_dw + p_o) * pi,
        end = pi + 2 * (p_dw + p_o + p_po) * pi
    ),
        linewidth = 0,
    alpha = 0.4, show.legend = FALSE) +
    geom_point(aes(
        x = x_duration,
        y = y_validity)) +
    scale_y_continuous(
        limits = c(30, 61),
        breaks = c(30, 40, 50, 60),
        labels = c("Trivially", "Simply", "Adequately", "Well"),
        name = "Representation of GCM") +
    scale_x_continuous(
        limits = c(30, 55),  # was 55 to  Keep original limits for the data
        breaks = c(30, 40, 50),
        name = "Time spent constructing model (minutes)") +
    scale_size(range = c(2, 7),
               breaks = range(gcm_metrics$n_adj_sets),
               name = "Adjustment\nsets") +
    scale_color_manual(values = c(hue1, hue2), name = "Is DAG?") +
    scale_fill_manual(values = c(hue1, hue2), guide = "none") +
    theme_minimal() +
    coord_fixed(ratio = 1, clip = "off") +
    theme(
        legend.position = "none",  # Hide legends from main plot
        # panel.grid.major.x = element_blank(),  # Remove vertical gridlines
        panel.grid.minor = element_blank(),
        plot.margin = margin(10, 10, 10, 10)
    )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

```r
# Create a dummy plot just to extract the legends
p_for_legend <- gcm_metrics |>
    mutate(
        x_duration = as.numeric(Modelling.duration, units = "minutes"),
        y_validity = GCM.validity * 10 + 20
    ) |>
    mutate(acyclic = if_else(acyclic, "Yes, DAG", "No, contained cycles")) |>
    ggplot(aes(
        x = x_duration,
        y = y_validity,
        color = acyclic,
        size = n_adj_sets
    )) +
    geom_point() +
    scale_size(range = c(2, 7),
               breaks = range(gcm_metrics$n_adj_sets),
               name = "N adjustment sets") +
    scale_color_manual(values = c(hue1, hue2), name = "Is DAG?") +
    theme_minimal() +
    theme(legend.box = "vertical")

# Extract the legend
legend_extracted <- cowplot::get_legend(p_for_legend)

# Custom legend plot showing the arc explanation
p_legend_arcs <- ggplot() +
    geom_arc_bar(aes(x0 = 0, y0 = 0, r0 = 0.1, r = 0.8, start = pi, end = pi + pi/3),
             fill = "black", alpha = 1, linewidth = 0) +
    geom_arc_bar(aes(x0 = 0, y0 = 0, r0 = 0.1, r = 0.6, start = pi + pi/3, end = pi + (2*pi/3
             fill = "black", alpha = 0.6, linewidth = 0) +
    geom_arc_bar(aes(x0 = 0, y0 = 0, r0 = 0.1, r = 0.4, start = pi + (2*pi/3), end = 2 * pi)
             fill = "black", alpha = 0.4, linewidth = 0) +
    annotate("text",
             x = 0.5, y = -0.9,
             label = "In data\nwarehouse",
             hjust = 0, size = 3, color = "black") +
    annotate("text",
             x = 0.5, y = 0,
             label = "Observable",
             hjust = 0, size = 3, color = adjustcolor("black", alpha.f = 0.8)) +
    annotate("text", x = 0.5, y = 0.9, label = "Potentially\nobservable",
             hjust = 0, size = 3, color = adjustcolor("black", alpha.f = 0.6)) +
```

```
    annotate("segment", x = 0.1, xend = 0.4, y = -0.5, yend = -0.6, color = "black") +
    annotate("segment", x = 0, xend = 0.4, y = 0, yend = 0,
             color = "black", alpha = 0.6) +
    annotate("segment", x = 0.1, xend = 0.4, y = 0.4, yend = 0.6,
             color = "black", alpha = 0.4) +
    coord_fixed() +
    xlim(-1, 2.5) +
    ylim(-1.3, 1.3) +
    labs(title = "Proportion of nodes:") +
    theme_void() +
    theme(plot.title = element_text(size = 9, face = "bold"))

# Combine the arc legend and extracted legends vertically
legend_column <- cowplot::plot_grid(
    legend_extracted,
    p_legend_arcs,
    ncol = 2,
    rel_heights = c(1, 1)
)

# Combine main plot with legend column
p_time_and_representation <- cowplot::plot_grid(
    p_main,
    legend_column,
    ncol = 2,
    rel_widths = c(1, 1)  # Adjust ratio as needed
)

p_time_and_representation
```




```
ggsave("figures/FIG--time-modelling-and-representation-arc-bars.jpg",
       plot = p_time_and_representation,
       width = 10, height = 5,
       dpi = 900)
```

As a table:

```
gcm_metrics |>
    mutate(acyclic = if_else(acyclic, "Yes, DAG", "No, contained cycles")) |>
    arrange(desc(Modelling.duration)) |>
    select(Modelling.duration, GCM.validity, acyclic, n_adj_sets, p_dw, p_o, p_po, p_l)
```

```
# A tibble: 8 x 8
  Modelling.duration GCM.validity acyclic   n_adj_sets   p_dw    p_o  p_po   p_l
  <Period>                  <dbl> <chr>          <int>  <dbl>  <dbl> <dbl> <dbl>
1 50M 23S                     4   Yes, DAG           1 0.111  0.167  0.278 0.444
2 50M 10S                     3.17 Yes, DAG          2 0.0909 0      0.364 0.545
3 46M 6S                      3.5  Yes, DAG          1 0.25   0      0.333 0.417
4 46M 4S                      3    Yes, DAG          1 0.133  0      0.533 0.333
5 41M 13S                     2.5  No, cont~         0 0.286  0      0.357 0.357
```

```
6 38M 55S                    2.5  Yes, DAG          6 0.0952 0.238  0.286 0.381
7 36M 37S                    3    No, cont~         0 0.188  0      0.312 0.5
8 34M 43S                    1.5  No, cont~         7 0.125  0.0625 0.188 0.625
```

Figure (OPTIONAL): Time spent with model and modelling utility - including graph similarity

Graph similarity computed with d-Measure.

Only showing the upper 50% of edges

```
d_between_models_median <- median(pull(d_edges, edge_d))

print(str_c("Median d-measure between models is ", round(d_between_models_median, 3)))
```

```
[1] "Median d-measure between models is 0.281"
```

```
d_edges |>
    slice(1)
```

```
# A tibble: 1 x 4
  from  to    edge_d edge_w
  <fct> <fct>  <dbl>  <dbl>
1 S3    S4     0.172   5.82
```

```
d_edges |>
    arrange(edge_w) |>
    slice(1)
```

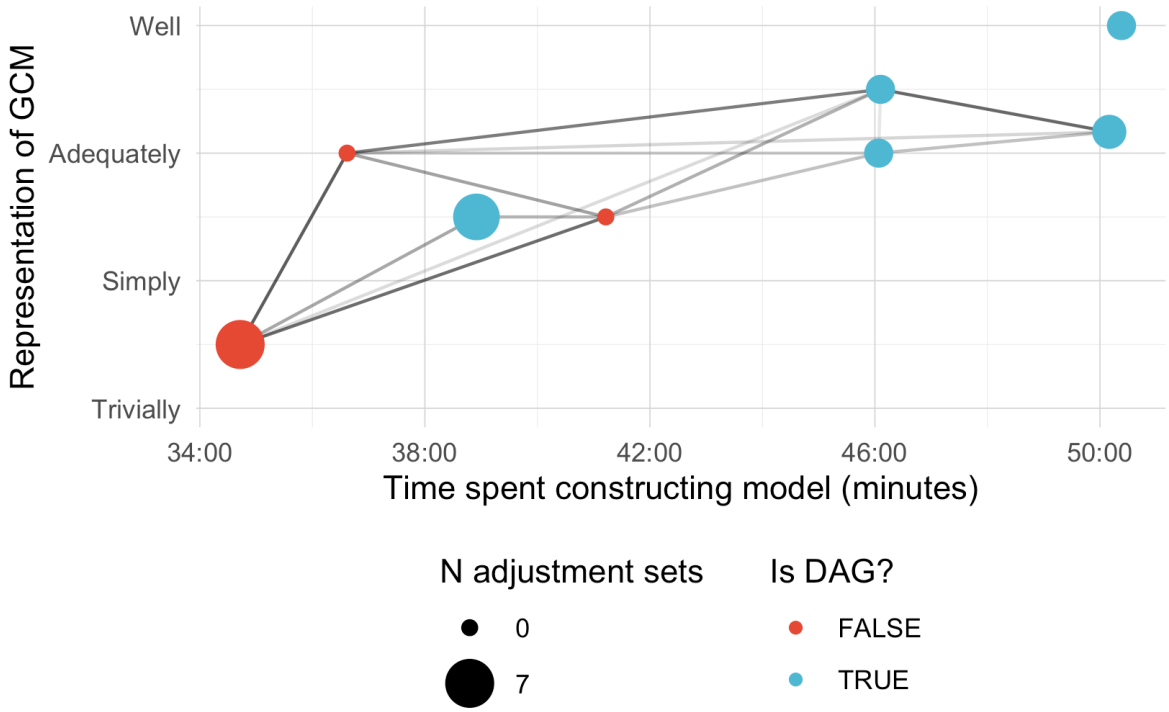
```
# A tibble: 1 x 4
  from  to    edge_d edge_w
  <fct> <fct>  <dbl>  <dbl>
1 S2    S5     0.475   2.10
```

```
d_between_models_by_time_and_utility <-
    d_edges |>
    inner_join(
        gcm_metrics |>
            mutate(from = Model) |>
            select(from, xfrom = Modelling.duration, yfrom = GCM.validity)
```

```
    ) |>
    inner_join(
        gcm_metrics |>
            mutate(to = Model) |>
            select(to, xto = Modelling.duration, yto = GCM.validity)
    )
```

```
Joining with `by = join_by(from)`
Joining with `by = join_by(to)`
```

```
p_time_and_representaion_with_distance <-
    gcm_metrics |>
    mutate(acyclic = if_else(acyclic, "Yes, DAG", "No, contained cycles")) |>
    ggplot() +
    geom_segment(
        data = d_between_models_by_time_and_utility |>
            filter(edge_d < d_between_models_median),
        aes(x = xfrom, xend = xto, y = yfrom, yend = yto,
            alpha = edge_w), size = 0.5, color = "black") +
    geom_point(data = gcm_metrics,
               aes(x = Modelling.duration, y = GCM.validity, size = n_adj_sets, colour = acy
    # geom_text(aes(label = Model), nudge_y = -.1) +
    scale_y_continuous(limits = c(1, 4),
                       breaks = c(1:4),
                       labels = c("Trivially", "Simply", "Adequately", "Well"),
                       name = "Representation of GCM") +
     scale_x_time(labels = label_time(format = "%M:%S"),
                  name = "Time spent constructing model (minutes)") +
    scale_size(range = c(2,7),
               breaks = range(gcm_metrics$n_adj_sets),
               # labels = c("None", )
               name = "N adjustment sets") +
    scale_alpha_continuous(range = c(0.1, 0.7), guide = "none") +
    scale_color_npg(name = "Is DAG?") +
    theme_minimal() +
    theme(legend.position = "bottom",
          legend.direction = "vertical",
          panel.grid.major = element_line(color = "grey85", linewidth = 0.2),
          panel.grid.minor = element_line(color = "grey92", linewidth = 0.1)
    )

p_time_and_representaion_with_distance
```

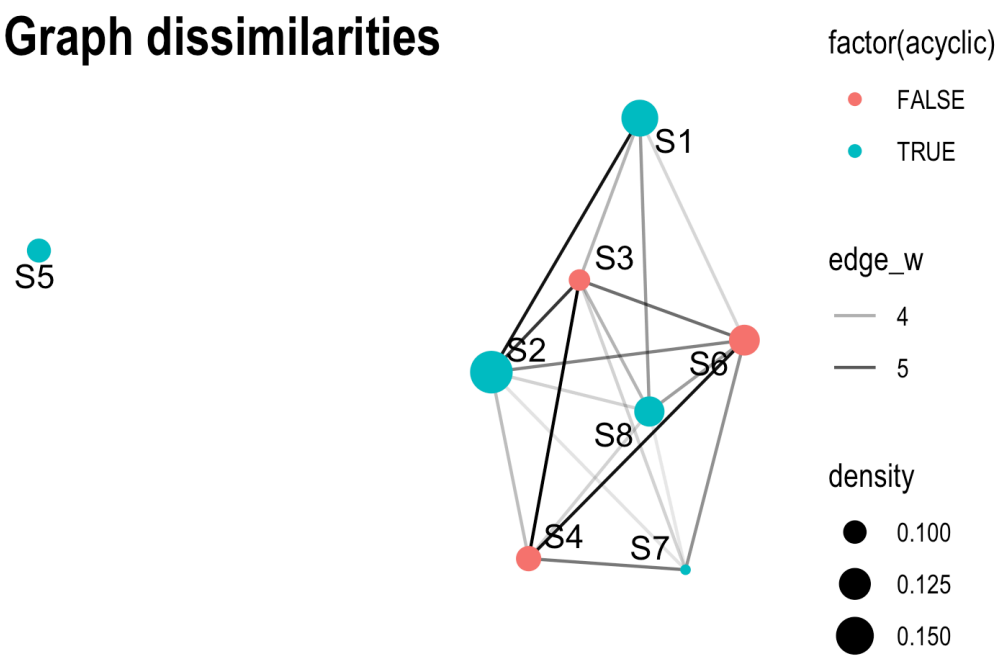



```
# ggsave("figures/FIG--time-modelling-and-representation-with-distances.jpg",
#        width = 5, height = 4,
#        dpi = 900)
```

DAG networks

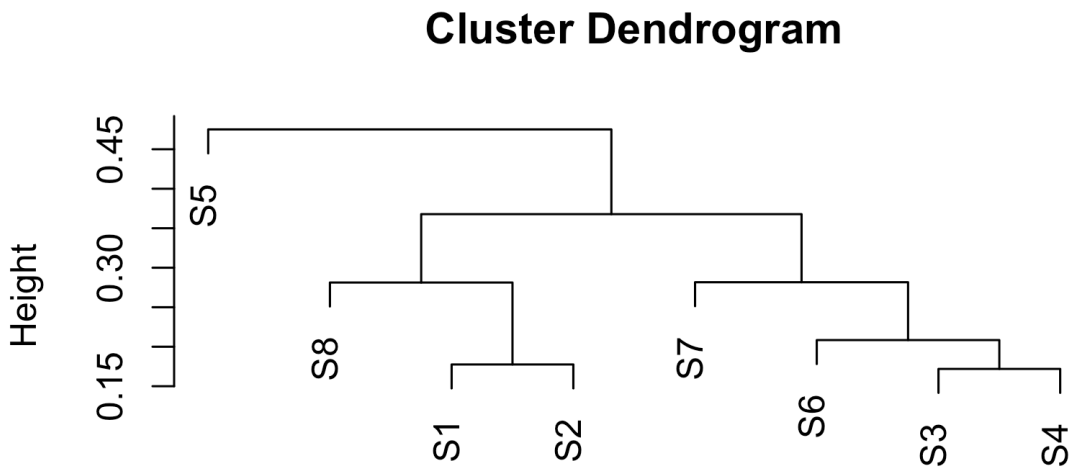
Visualising model dissimilarities

```
tbl_graph(
    nodes = dag_complexity_metrics |> rename(name = Model),
    edges = d_edges) |>
    activate("edges") |>
    filter(edge_d < 0.33) |>
    ggraph(layout = "fr") +
    geom_edge_fan(aes(alpha = edge_w)) +
    geom_node_point(aes(size = density, color = factor(acyclic))) +
    geom_node_text(aes(label = name), repel = T) +
    theme_graph() +
    ggtitle("Graph dissimilarities")
```



Same again, but hierarchical clustering

```
hc <- hclust(d_matrix |> select(-from) |> as.dist())
plot(hc)
```



Edge coherence

```
tb_edge_coherence <-
    edges_with_metrics |>
    summarise(
        n = sum(n),
        n_desc = sum(n_desc),
        max_n = sum(max_n),
        aligned = mean(switched == "aligned"),
        contested = mean(switched != "aligned")
    ) |>
    mutate(edges_included = n / max_n,
           desc_incllude = n_desc / max_n)
tb_edge_coherence
```

```
# A tibble: 1 x 7
      n n_desc max_n aligned contested edges_included desc_incllude
  <int>  <int> <int>   <dbl>     <dbl>          <dbl>         <dbl>
1   204    258   258   0.938    0.0622          0.791             1
```

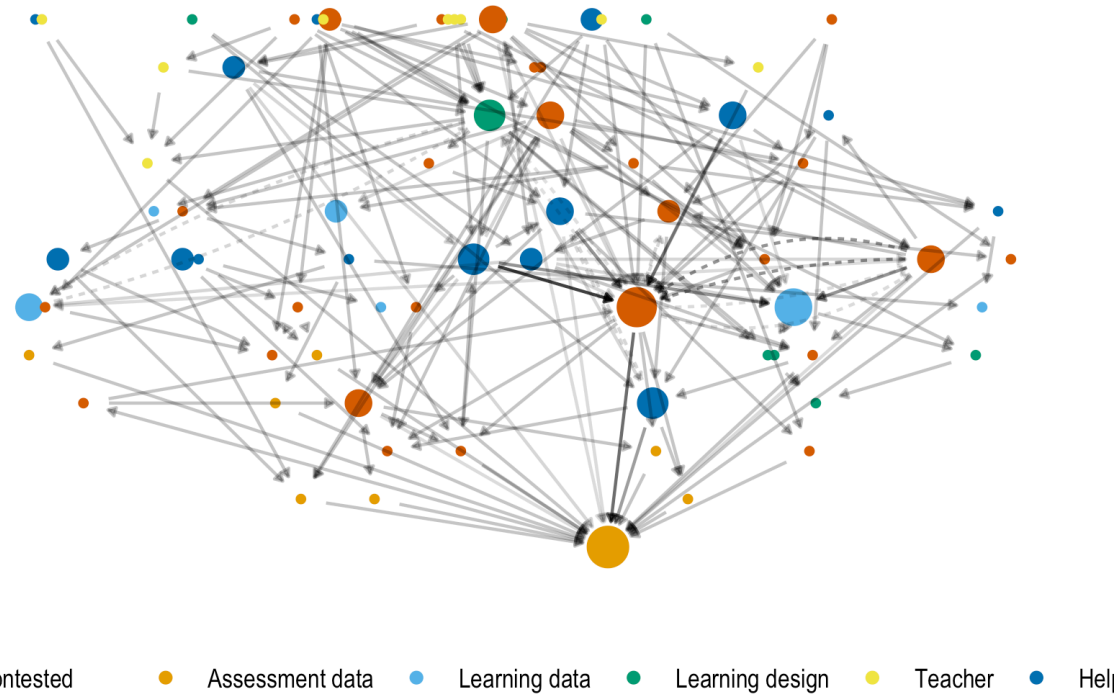
Generally when two nodes in the model were selected by different participants the same edge was drawn in 79.07% of the time. A handful of potential edges (6.22%) were included, but asigned a different direction of causal flow. In 100% of the cases where a path $A \rightarrow B$ was included in a model, then there existed a path from A to B in every other model that included those two nodes, just sometimes through a mediator(s) variable.

Merging graphs

```
merged_all_graph_layout <- create_layout(merged_all_graph, layout = "sugiyama")
merged_all_graph_layout_adjusted <-
    merged_all_graph_layout |>
    mutate(
        # x = if_else(name == "S.Post.Fb",x + 20,x),
        y = if_else(name == "S.Act.Fb.Post",y + 4,y)
           ) |>
    mutate(
        x = if_else(name == "S.Act",x + 20,x),
        y = if_else(name == "S.Act",y - 4,y)
           )
ptemp <- merged_all_graph_layout_adjusted |>
    ggraph() +
        geom_edge_fan(aes(alpha = edge_w), strength = 0.2,
            arrow = arrow(length = unit(2, 'mm'), type = "closed"),
            start_cap = circle(3, 'mm'),
            end_cap = circle(3, 'mm')) +
        geom_node_point(aes(colour = node_type, size = node_w)) +
        # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.3,1)) +
        theme(legend.position = "none")

b <- ggplot_build(ptemp)
x_limits_fine <- b$layout$panel_params[[1]]$x.range
y_limits_fine <- b$layout$panel_params[[1]]$y.range



plot_fine_graph_from_layout <- function(lyt, weight_nodes = TRUE) {
    ppp <-
        lyt |>
```

```
        ggraph()

    if (weight_nodes) {
        pp <- ppp +
            geom_node_point(aes(colour = node_type,
                                size = node_w)) +
            scale_size(name = "point-size") +
            geom_edge_fan(aes(
                alpha = edge_p_w * edge_w + edge_w,
                edge_linetype = factor(switched)),
                strength = 0.2,
                arrow = arrow(length = unit(1, 'mm'),
                              type = "closed"),
                start_cap = circle(3, 'mm'),
                end_cap = circle(3, 'mm'))
    } else {
        pp <- ppp +
            geom_node_point(aes(colour = node_type)) +
            geom_edge_fan(strength = 0.2, alpha = 0.5,
                          arrow = arrow(length = unit(1, 'mm'),
                                        type = "closed"),
                          start_cap = circle(1, 'mm'),
                          end_cap = circle(1, 'mm'))

    }
    pp +
        scale_colour_okabe_ito() +
        # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.3,1)) +
        theme(legend.position = "none",
              plot.margin = margin(t = 0, r = 0, b = 0, l = 0, unit = "pt")) +
        xlim(x_limits_fine) +
        ylim(y_limits_fine)
}

g_plot_fine <- function(g, lyt = merged_all_graph_layout_adjusted, weight_nodes = TRUE) {
    create_layout(g, layout = "sugiyama") |>
        left_join(lyt |> select(x,y,name,node_group,node_type,node_w), by = "name",
                  suffix = c(".default", "")) |>
        plot_fine_graph_from_layout(weight_nodes = weight_nodes)
```

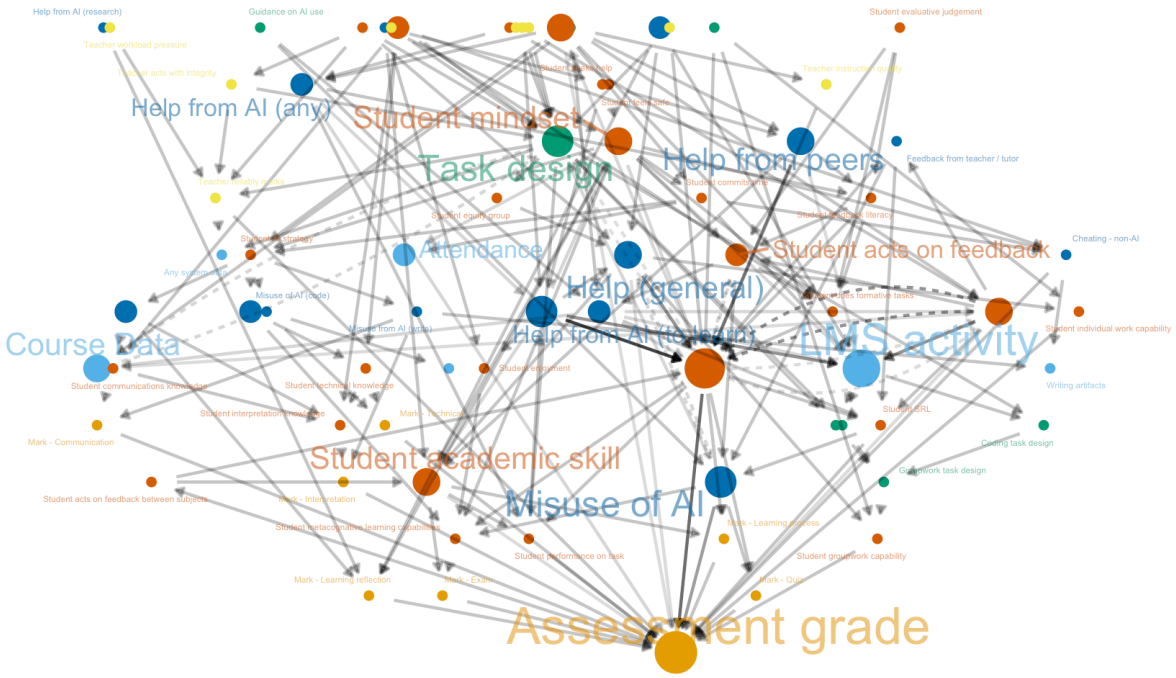
```
}

pMergedFineLegend <- merged_all_graph_layout_adjusted |>
    g_plot_fine() +
    theme(legend.position = "bottom") +
    guides(size = "none", edge_width = "none", alpha = "none", edge_alpha = "none",
           color = guide_legend(nrow = 1)) +
    theme(legend.title = element_blank())
pMergedFineLegend
```




```
pMergedFineLegendTall <- merged_all_graph_layout_adjusted |>
    g_plot_fine() +
    theme(legend.position = "right") +
    guides(size = "none", alpha = "none", edge_alpha = "none", edge_width = "none",
           color = guide_legend(ncol = 2)) +
    theme(legend.title = element_blank())
pMergedFine <- pMergedFineLegend +
    theme(legend.position = "none")

pMergedFineLabelled <- pMergedFine +
```

```
    geom_node_text(
        aes(label = nice_name, size = node_w, colour = node_type), alpha = 0.6,
        data = merged_all_graph_layout_adjusted |>  filter(node_w > 0), repel = TRUE, max.ove
    ggnewscale::new_scale("size") +
    scale_size(range = c(3,6), name = "text-size")
pMergedFineLabelled
```

```
Warning: ggrepel: 22 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```
# Extract legend
leg <- cowplot::get_legend(pMergedFineLegend)
```

```
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
```

```
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
```

```
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_stringMetric, as.graphicsAnnot(x$label)): font family
'Arial Narrow' not found in PostScript font database

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
```

```
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
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family 'Arial Narrow' not found in PostScript font database
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family 'Arial Narrow' not found in PostScript font database
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family 'Arial Narrow' not found in PostScript font database
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family 'Arial Narrow' not found in PostScript font database
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family 'Arial Narrow' not found in PostScript font database
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family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
```

```r
# Turn it into a ggplot object for saving
legplot <- cowplot::ggdraw(leg)
# Save just the legend
ggsave("dag_legend.jpg", legplot, width = 6, height = 1)

leg_tall <- cowplot::get_legend(pMergedFineLegendTall)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
```

```
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, : font
family 'Arial Narrow' not found in PostScript font database
```

```r
leg_tall_plot <- cowplot::ggdraw(leg_tall)


# ggsave("dag_merged_fine.svg", pMergedFine, width = 6, height = 4)
# ggsave("dag_merged_fine_labelled.svg", pMergedFineLabelled, width = 6, height = 4)
```

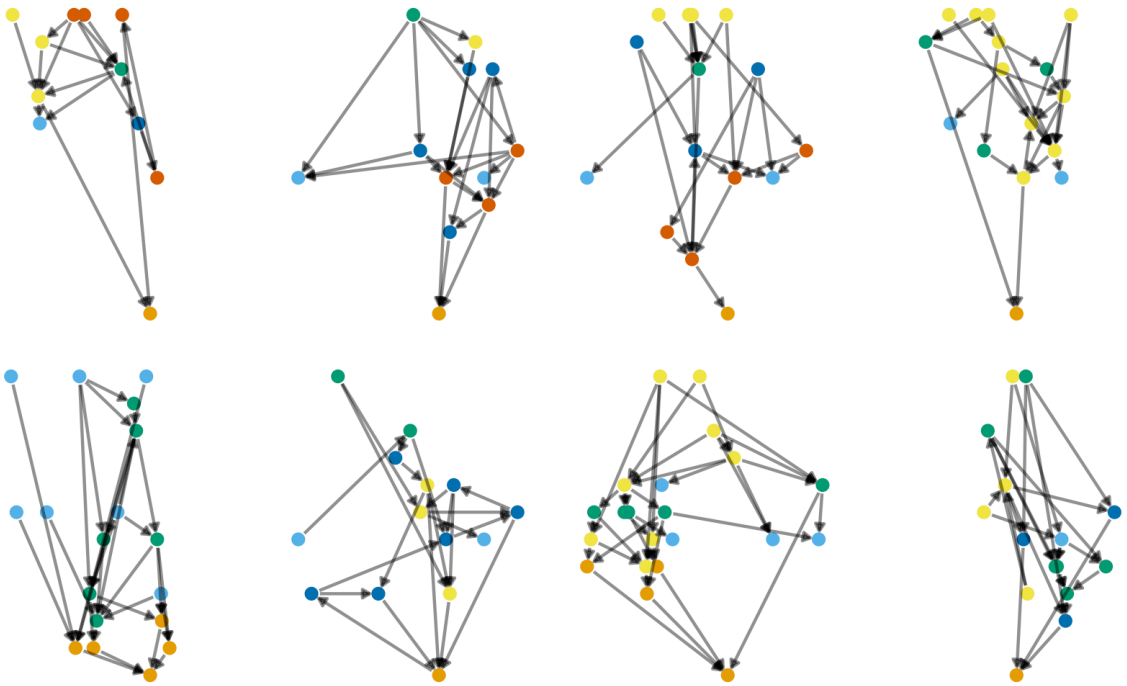
```r
p1 <- g_plot_fine(tdag_s1, weight_nodes = FALSE)
p2 <- g_plot_fine(tdag_s2, weight_nodes = FALSE)
p3 <- g_plot_fine(tdag_s3, weight_nodes = FALSE)
p4 <- g_plot_fine(tdag_s4, weight_nodes = FALSE)
```

```
p5 <- g_plot_fine(tdag_s5, weight_nodes = FALSE)
p6 <- g_plot_fine(tdag_s6, weight_nodes = FALSE)
p7 <- g_plot_fine(tdag_s7, weight_nodes = FALSE)
p8 <- g_plot_fine(tdag_s8, weight_nodes = FALSE)

# Plot border
    # theme(
    #   plot.background = element_rect(
    #     colour = "grey70",
    #     fill = NA,
    #     linewidth = 1
    #   )
    # )

hl_text_size <- 1.5
hl_text_alpha <- 0.7

p2peerHighlight <- p2 +
    geom_node_point(size = 3, shape = 1, alpha = 0.6,
    data = p2$data |>  filter(name == "H.Peer")) +
    geom_node_point(size = 5, shape = 1, alpha = 0.3,
    data = p2$data |>  filter(name == "H.Peer")) +
    geom_node_point(size = 7, shape = 1, alpha = 0.1,
    data = p2$data |>  filter(name == "H.Peer")) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   size = hl_text_size , alpha = hl_text_alpha,
    data = p2$data |>  filter(name == "H.Peer") |>
        inner_join(name_node_nice_name, by = "name"), repel = T)
p3peerHighlight <- p3 +
    geom_node_point(size = 3, shape = 1, alpha = 0.6,
    data = p3$data |>  filter(name == "H.Peer")) +
    geom_node_point(size = 5, shape = 1, alpha = 0.3,
    data = p3$data |>  filter(name == "H.Peer")) +
    geom_node_point(size = 7, shape = 1, alpha = 0.1,
    data = p3$data |>  filter(name == "H.Peer")) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   size = hl_text_size , alpha = hl_text_alpha,
    data = p3$data |>  filter(name == "H.Peer") |>
        inner_join(name_node_nice_name, by = "name"), repel = T)
p4peerHighlight <- p4 +
    geom_node_point(size = 3, shape = 1, alpha = 0.6,
    data = p4$data |>  filter(name == "H.Peer")) +
```

```
    geom_node_point(size = 5, shape = 1, alpha = 0.3,
    data = p4$data |>  filter(name == "H.Peer")) +
    geom_node_point(size = 7, shape = 1, alpha = 0.1,
    data = p4$data |>  filter(name == "H.Peer")) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   size = hl_text_size , alpha = hl_text_alpha,
    data = p4$data |>  filter(name == "H.Peer") |>
        inner_join(name_node_nice_name, by = "name"), repel = T)

p7KnHighlight <- p7 +
    geom_node_point(size = 3, shape = 5, alpha = 0.6,
    data = p7$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_point(size = 5, shape = 5, alpha = 0.3,
    data = p7$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_point(size = 7, shape = 5, alpha = 0.1,
    data = p7$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   size = hl_text_size , alpha = hl_text_alpha,
    data = p7$data |>  filter(str_detect(name, "Kn")) |>
        inner_join(name_node_nice_name, by = "name"), repel = T)

p5KnHighlight <- p5 +
    geom_node_point(size = 3, shape = 5, alpha = 0.6,
    data = p5$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_point(size = 5, shape = 5, alpha = 0.3,
    data = p5$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_point(size = 7, shape = 5, alpha = 0.1,
    data = p5$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   size = hl_text_size , alpha = hl_text_alpha,
    data = p5$data |>  filter(str_detect(name, "Kn")) |>
        inner_join(name_node_nice_name, by = "name"), repel = T)

p6KnHighlight <- p6 +
    geom_node_point(size = 3, shape = 5, alpha = 0.6,
    data = p6$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_point(size = 5, shape = 5, alpha = 0.3,
    data = p6$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_point(size = 7, shape = 5, alpha = 0.1,
    data = p6$data |>  filter(str_detect(name, "Kn"))) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   size = hl_text_size , alpha = hl_text_alpha,
```

```
    data = p6$data |>  filter(str_detect(name, "Kn")) |>
        inner_join(name_node_nice_name, by = "name"), repel = T)
#
```

```
pAllFine <- (p1 | p2 | p3 | p4) / (p5 | p6 | p7 | p8)
pAllFine
```



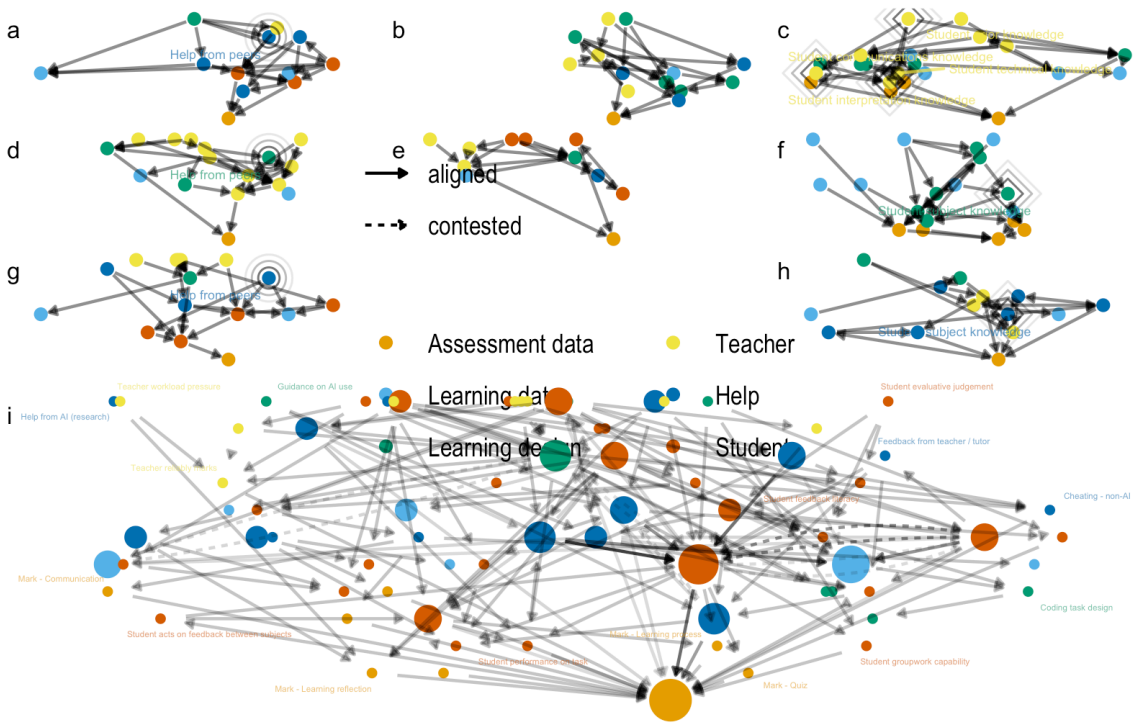
```
# ggsave("dags_All_fine.svg", pAllFine, width = 18, height = 9)
```

Figure: Model graphs, all and merged

```
library(grid)
tag_plot <- function(p, tag, x = 0, y = 0.9, fontsize = 8) {
  p + annotation_custom(
    grid::textGrob(tag, x = x, y = y,
                   hjust = 0, vjust = 1, gp = gpar(fontsize = fontsize))
  )
}
```

```
pAll3by3 <-
    (
        tag_plot(p2peerHighlight,"a")|
            tag_plot(p8, "b") |
            tag_plot(p7KnHighlight, "c")) /
    (
        tag_plot(p4peerHighlight, "d") |
            tag_plot(p1, "e") |
            tag_plot(p5KnHighlight, "f")) /
    (
        tag_plot(p3peerHighlight, "g") |
            leg_tall_plot |
            tag_plot(p6KnHighlight, "h"))


pFineTogetherLabelled <- pAll3by3 / tag_plot(pMergedFineLabelled, "i") + plot_layout(heights
pFineTogether <- pAll3by3 / pMergedFine + plot_layout(heights = c(1,1,1,3))
pFineTogetherLabelled
```

```
Warning: ggrepel: 58 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```
# ggsave("dags_fine_all_and_merged.png", pFineTogether, width = 9, height = 10, dpi = 900)
# ggsave("dags_fine_all_and_merged.svg", pFineTogether, width = 9, height = 10)
ggsave("figures/FIG--dags_fine_all_and_merged_labelled.jpg", pFineTogetherLabelled, width = 9
```

```
Warning: ggrepel: 14 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

```
# ggsave("dags_fine_all_and_merged_labelled.svg", pFineTogetherLabelled, width = 9, height =

print("a, d, g are models S2, S3, S4; b, e are S8 and S1; c, f, h are S7, S5, S6. Closest mo
```

```
[1] "a, d, g are models S2, S3, S4; b, e are S8 and S1; c, f, h are S7, S5, S6. Closest model
```

Common patterns - fine

Most common three-node patterns.

```
merged_all_graph |>
    activate("nodes") |>
    filter(node_w > 3) |>
    activate("edges") |>
    filter(edge_w > 1) |>
    g_plot_fine() +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   alpha = hl_text_alpha, repel = TRUE)
```

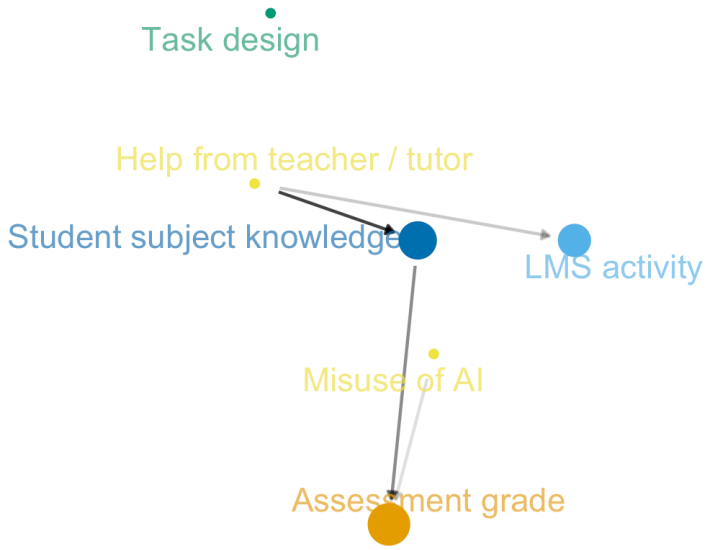


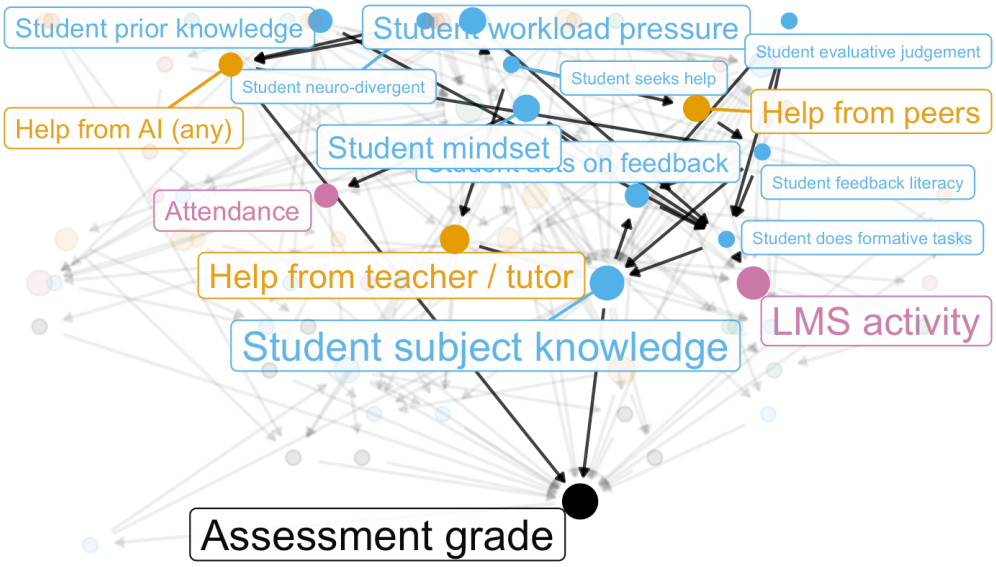
Figure - all graphs, single highlight

```
plot_highlighted_fine_graph <- function(model = 1, LEGEND_POSITION = "none") {
    MODEL <- paste0("s", model)
    nodes_in_graph <- all_nodes |>
        filter(Model == MODEL) |>
        pull(name)
    edges_in_graph <- all_edges |>
        filter(Model == MODEL) |>
```

```
    select(from, to)

edges_in_graph_indexed <-
    edges_in_graph |>
    mutate(
        from = match(from, merged_all_graph %N>% as_tibble() |> pull(name)),
        to = match(to, merged_all_graph %N>% as_tibble() |> pull(name))
    )

g_highlight <- merged_all_graph %N>%
    mutate(
        node_highlight = if_else(
            name %in% nodes_in_graph, 2, 1
        )) %E>%
    left_join(
        edges_in_graph_indexed |>
            mutate(edge_highlight = 2)
    ) |>
    mutate(edge_highlight = replace_na(edge_highlight, 1)) |>
    create_layout("sugiyama")

g_highlight |>
    ggraph() +
    geom_edge_fan(aes(
        # edge_linetype = factor(switched),
        alpha = edge_highlight),
        strength = 0.2,
        arrow = arrow(length = unit(1, 'mm'),
                      type = "closed"),
        start_cap = circle(3, 'mm'),
        end_cap = circle(3, 'mm')) +
    geom_node_label(
        aes(label = nice_name,
            color = node_type,
            size = node_w),
        alpha = 0.95,
        data = g_highlight |>
            filter(node_highlight == 2),
        repel = T, max.overlaps = 30,
        show.legend = FALSE) +
    geom_node_point(aes(colour = node_type,
                        size = node_w,
```

```
                              alpha = node_highlight)) +
        scale_size(name = "point-size", range = c(2,5)) +
        scale_color_okabe_ito(order = c(9, 7,1, 2,3, 6, 8,5, 4)) +
        # scale_color_npg() +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.08,0.8)) +
        scale_alpha_continuous(range = c(0.1,1.0)) +
        # theme(legend.position = "none",
        #       plot.margin = margin(t = 0, r = 0, b = 0, l = 0, unit = "pt")) +
        theme(legend.position = LEGEND_POSITION) +
        guides(size = "none", edge_width = "none", alpha = "none", edge_alpha = "none",
               color = guide_legend(ncol = 1)) +
        theme(legend.title = element_blank())
}

plot_highlighted_fine_graph(4)
```

```
Joining with `by = join_by(from, to)`
```



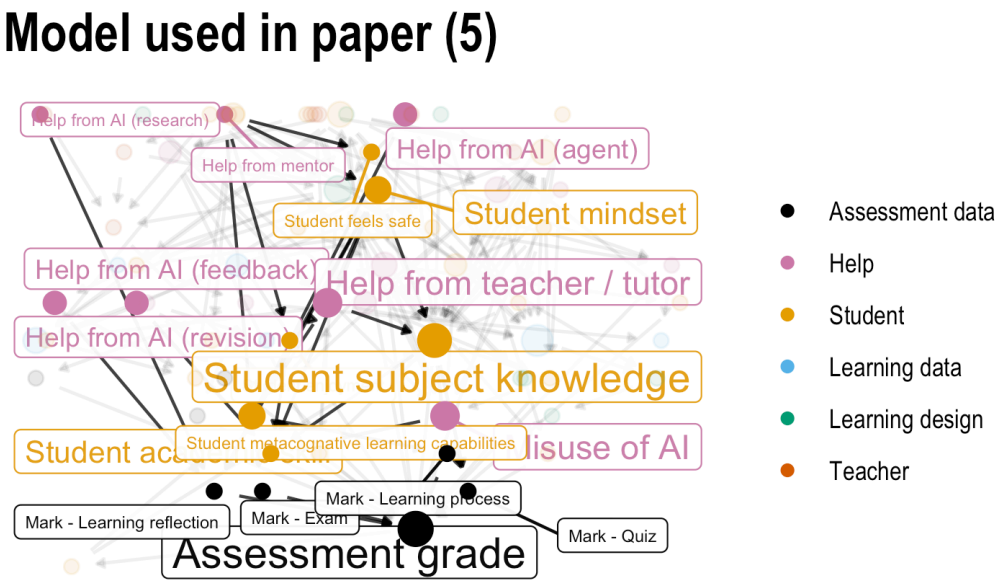
```
# +
#     xlim(x_limits_fine) +
#     ylim(y_limits_fine)



plot_highlighted_coarse_graph <- function(model = 1, LEGEND_POSITION = "none") {
    MODEL <- paste0("s", model)
    node_groups_in_graph <- all_nodes |>
        filter(Model == MODEL) |>
        mutate(name = code_node_group(name)) |>
        distinct(name) |>
        pull(name)
    edges_in_coarse_graph <- all_edges |>
        filter(Model == MODEL) |>
        select(from, to) |>
        mutate(across(from:to, code_node_group))

    edges_in_graph_indexed <-
        edges_in_coarse_graph |>
        mutate(
            from = match(from, merged_coarse_models %N>% as_tibble() |> pull(name)),
            to = match(to, merged_coarse_models %N>% as_tibble() |> pull(name))
        )

    edge_integers <-
        edges_in_graph_indexed |>
        mutate(p2p3 = (2 ** from) * (3 ** to) ) |>
        pull(p2p3)

    node_types <- merged_all_graph %N>%
        as_tibble() |>
        distinct(node_group, node_type)

    g_highlight <- merged_coarse_models %N>%
        inner_join(node_types |> rename(name = node_group),
                   by = "name") |>
        mutate(
            node_highlight = if_else(
                name %in% node_groups_in_graph, 2, 1
            )) %E>%
        mutate(edge_highlight = if_else(
```

```
            ((2 ** from )*(3 ** to)) %in% edge_integers, 2, 1
        )) |>
        create_layout("sugiyama")

    g_highlight |>
        ggraph() +
        geom_edge_fan(aes(
            # edge_linetype = factor(switched),
            alpha = edge_highlight),
            strength = 0.2,
            arrow = arrow(length = unit(1, 'mm'),
                          type = "closed"),
            start_cap = circle(3, 'mm'),
            end_cap = circle(3, 'mm')) +
        geom_node_label(
            aes(label = nice_name,
                color = node_type,
                size = node_w),
            alpha = 0.95,
            data = g_highlight |>
                filter(node_highlight == 2),
            repel = T, max.overlaps = 30,
            show.legend = FALSE) +
        geom_node_point(aes(colour = node_type,
                            size = node_w,
                            alpha = node_highlight)) +
        scale_size(name = "point-size", range = c(2,5)) +
        # scale_color_npg() +
        scale_color_okabe_ito(order = c(9, 7,1, 2,3, 6, 8,5, 4)) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.08,0.8)) +
        scale_alpha_continuous(range = c(0.1,1.0)) +
        # theme(legend.position = "none",
        #       plot.margin = margin(t = 0, r = 0, b = 0, l = 0, unit = "pt")) +
        theme(legend.position = LEGEND_POSITION) +
        guides(size = "none", edge_width = "none", alpha = "none", edge_alpha = "none",
               label = "none",
               color = guide_legend(ncol = 1)) +
        theme(legend.title = element_blank())
}

# Model used in paper
```

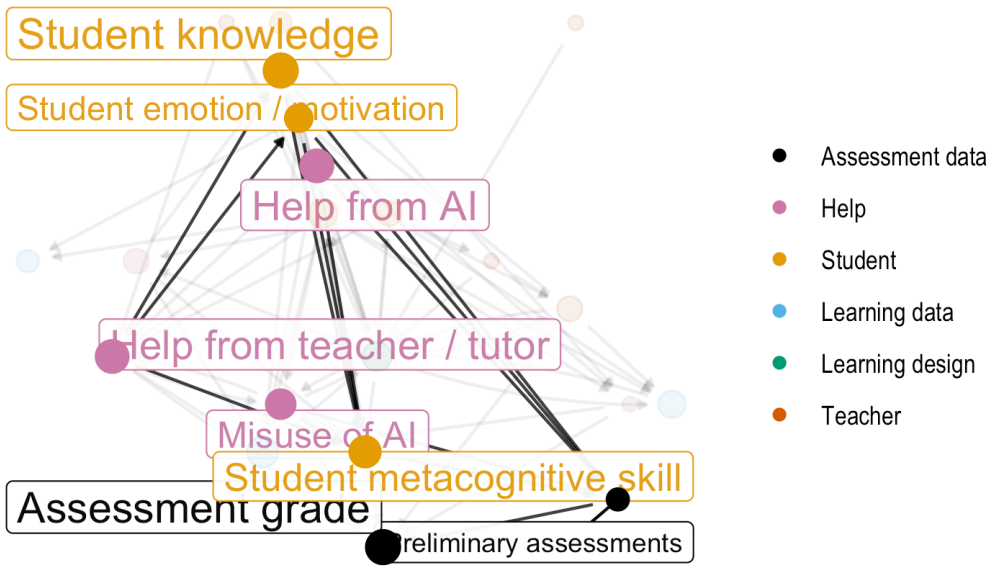
```
p_merged_highlighted_fine <- plot_highlighted_fine_graph(5, LEGEND_POSITION = "right")
```

```
Joining with `by = join_by(from, to)`
```

```
p_merged_highlighted_coarse <- plot_highlighted_coarse_graph(5, LEGEND_POSITION = "right")
p_merged_highlighted_fine + ggtitle("Model used in paper (5)")
```




```
p_merged_highlighted_coarse
```




```
ggsave(plot = p_merged_highlighted_fine,
       filename = "figures/FIG--merged-fine-with-1-highlighted.jpg",
       width = 8, height = 5, dpi = 600)

ggsave(plot = p_merged_highlighted_coarse,
       filename = "figures/FIG--merged-coarse-with-1-highlighted.jpg",
       width = 8, height = 5, dpi = 600)
```

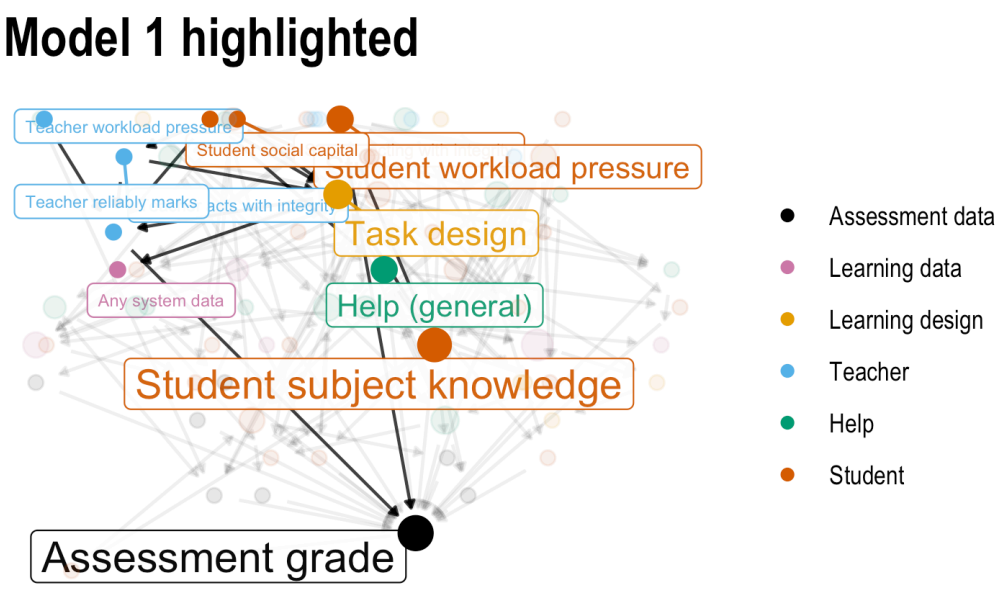
```
# Generating all
p_fine_show_and_save <- function(n){
    p_temp <- plot_highlighted_fine_graph(n, LEGEND_POSITION = "right") + ggtitle(paste0("Mo
    ggsave(plot = p_temp,
           filename = paste0("figures/plt--merged-fine-with-", n, "-highlighted.jpg"),
           width = 8, height = 5, dpi = 600)
    return(p_temp)
}
p_coarse_show_and_save <- function(n){
    p_temp <- plot_highlighted_coarse_graph(n, LEGEND_POSITION = "right") + ggtitle(paste0("
    ggsave(plot = p_temp,
           filename = paste0("figures/plt--merged-coarse-with-", n, "-highlighted.jpg"),
```

```
            width = 8, height = 5, dpi = 600)
    return(p_temp)
}
```

Highlighting all models in turn

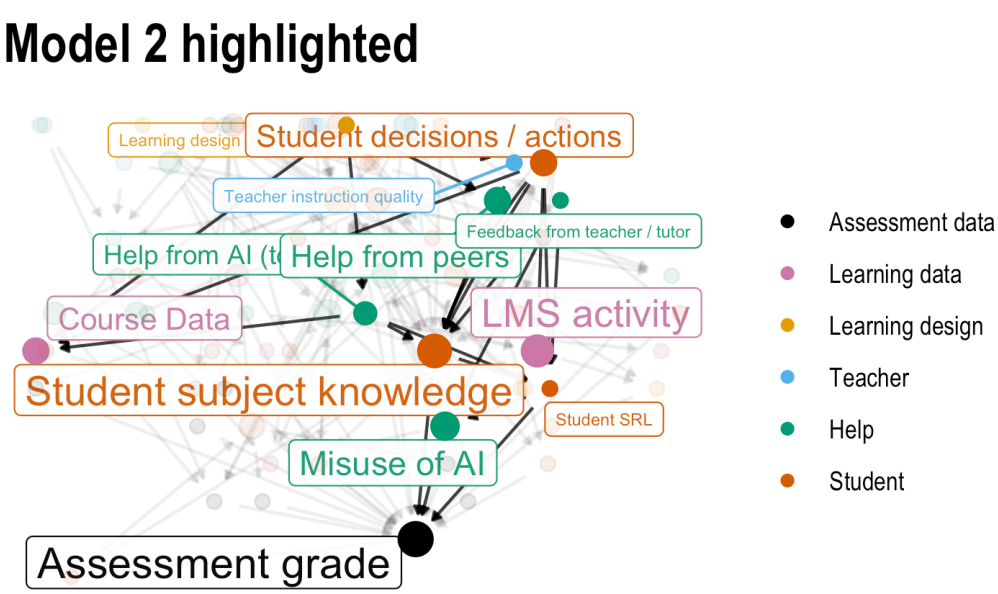
```
p_fine_show_and_save(1)
```

```
Joining with `by = join_by(from, to)`
```



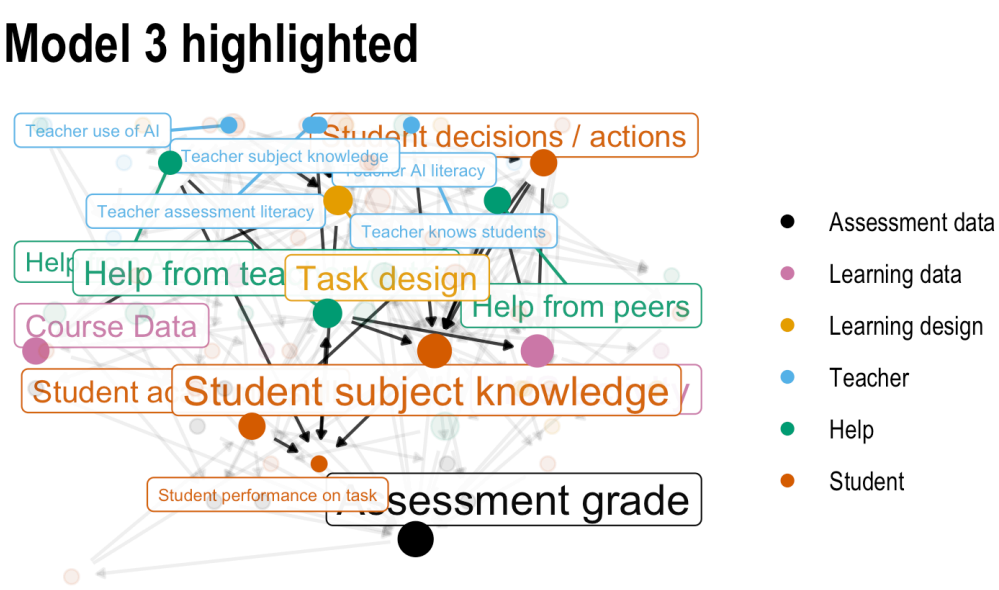

```
p_fine_show_and_save(2)
```

```
Joining with `by = join_by(from, to)`
```




```
p_fine_show_and_save(3)
```

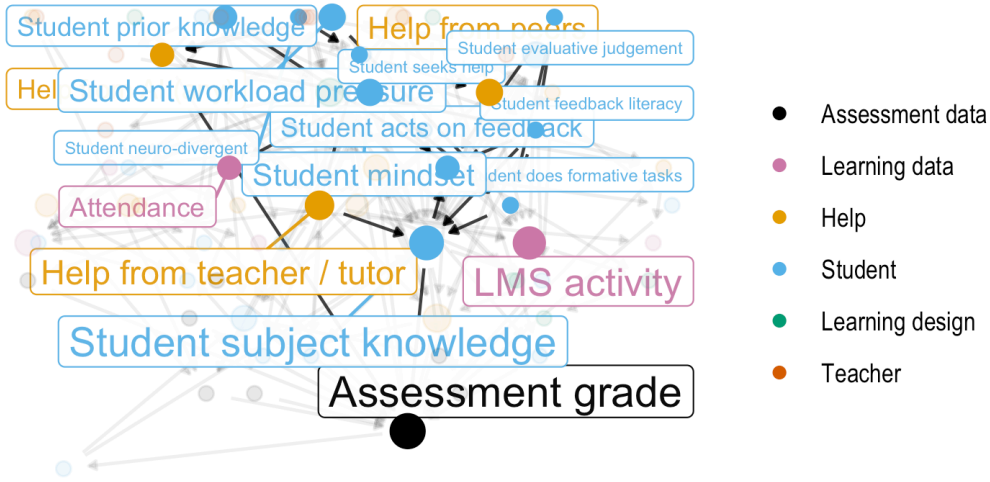
```
Joining with `by = join_by(from, to)`
```




```
p_fine_show_and_save(4)
```

```
Joining with `by = join_by(from, to)`
```

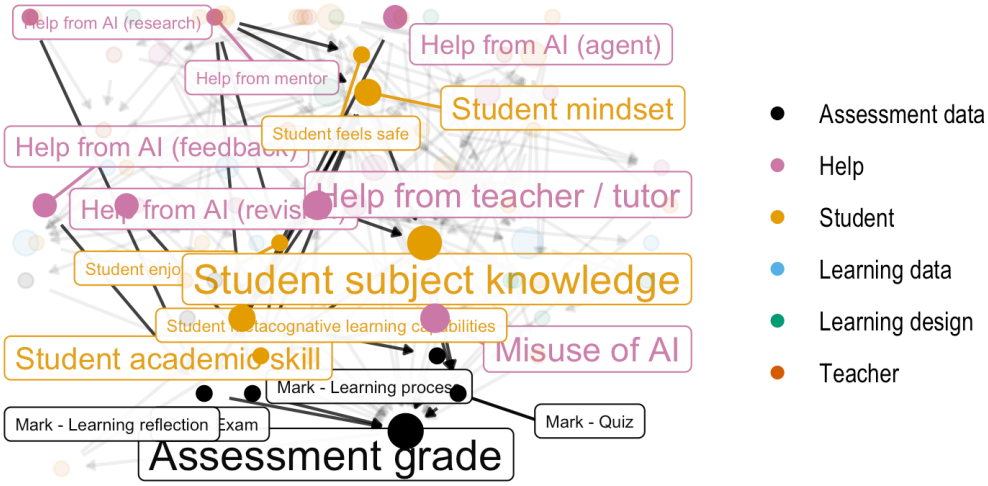
Model 4 highlighted




```
p_fine_show_and_save(5)
```

```
Joining with `by = join_by(from, to)`
```

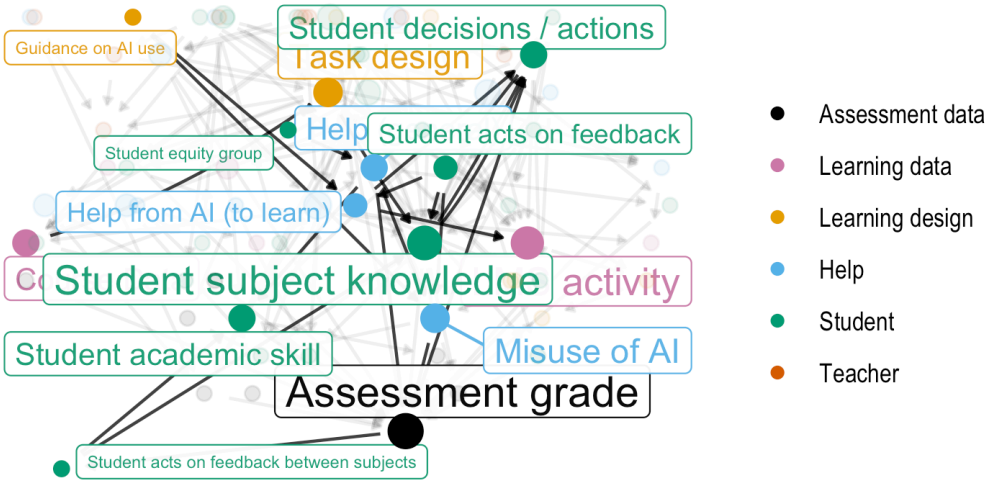
Model 5 highlighted




```
p_fine_show_and_save(6)
```

```
Joining with `by = join_by(from, to)`
```

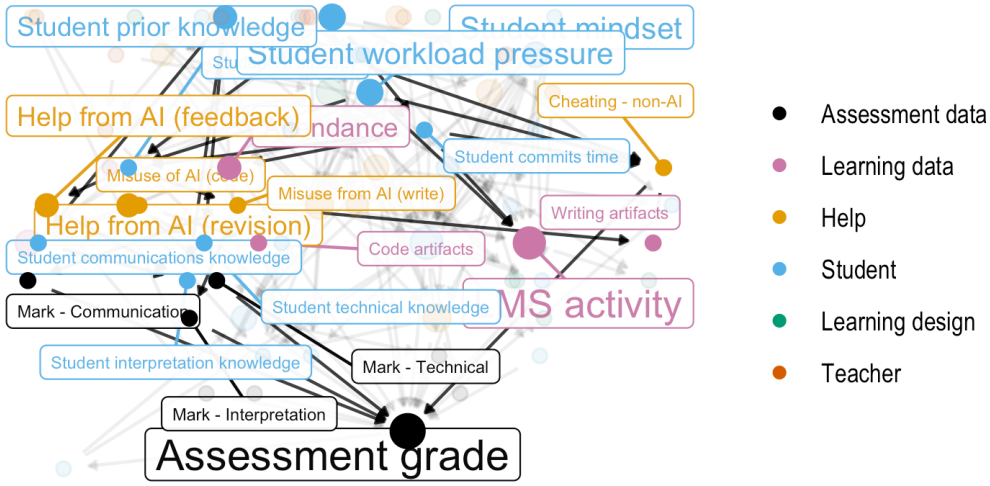
Model 6 highlighted




```
p_fine_show_and_save(7)
```

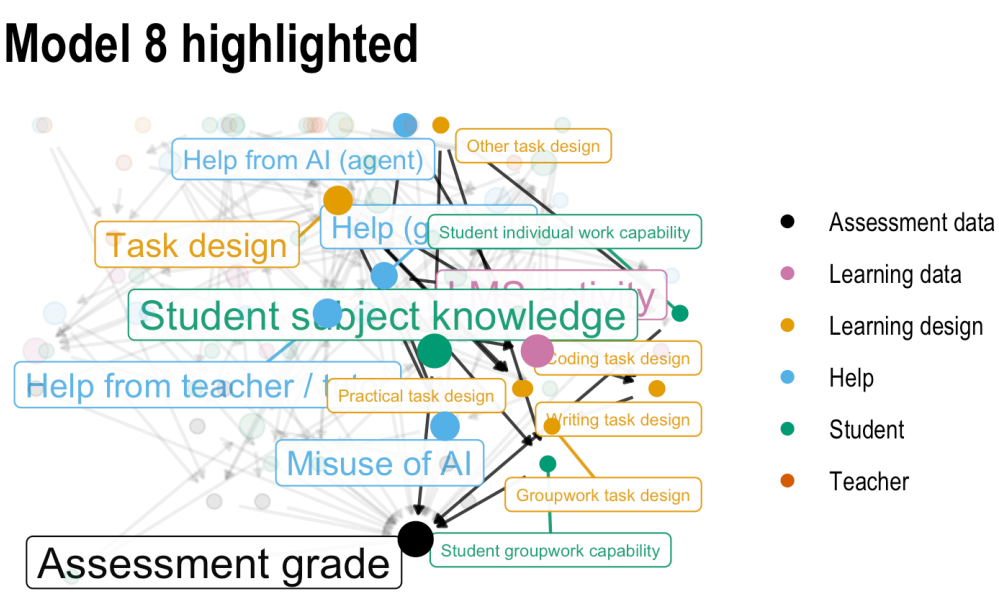
```
Joining with `by = join_by(from, to)`
```

Model 7 highlighted



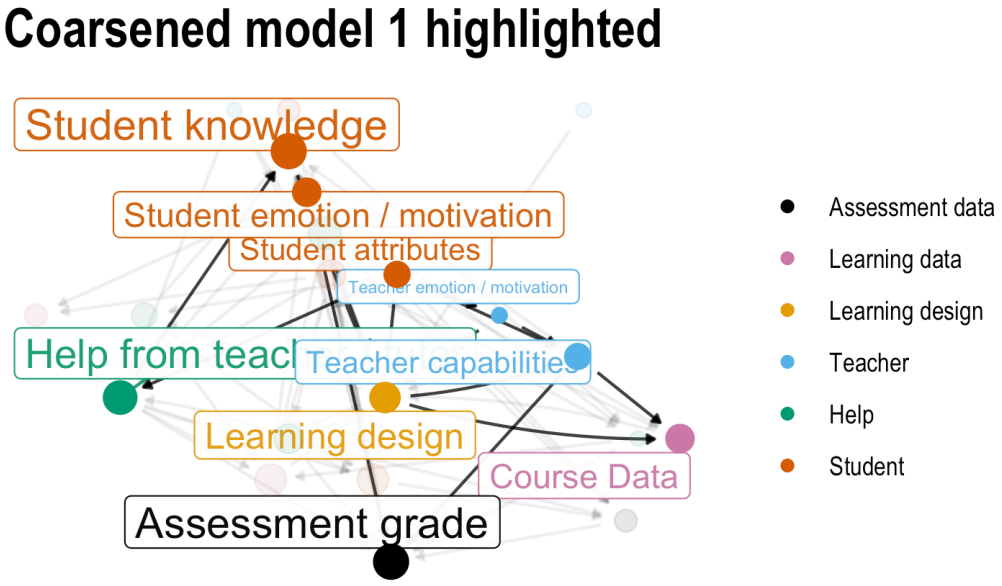

```
p_fine_show_and_save(8)
```

```
Joining with `by = join_by(from, to)`
```

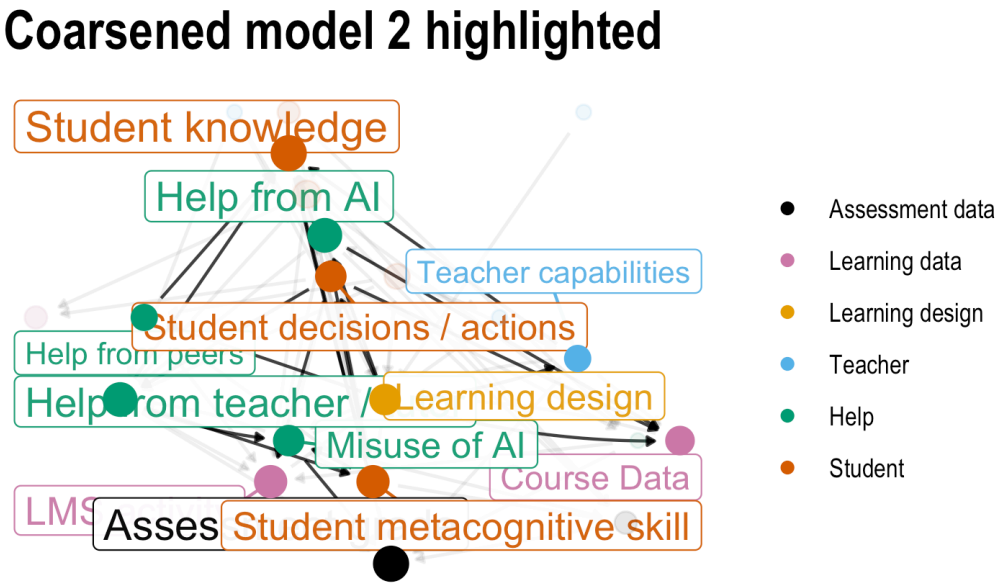


Highlighting all coarsened models in turn

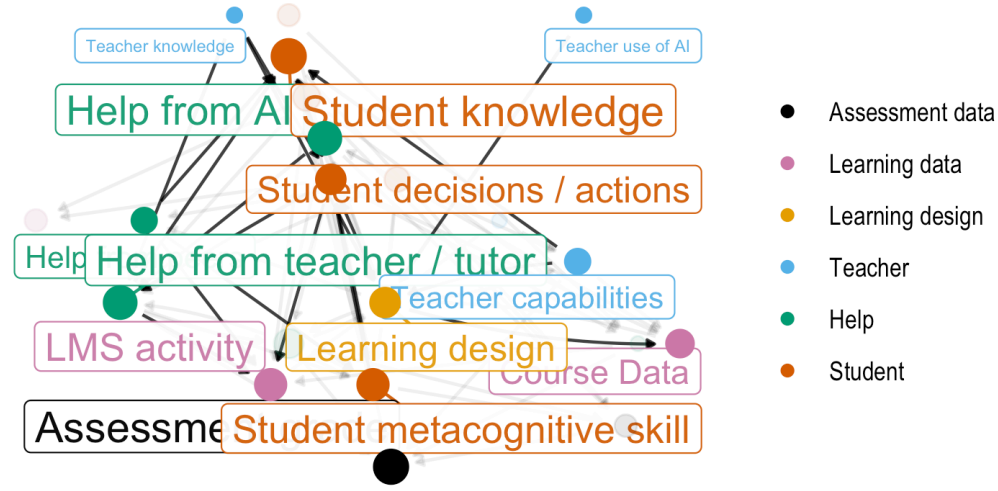
```
p_coarse_show_and_save(1)
```



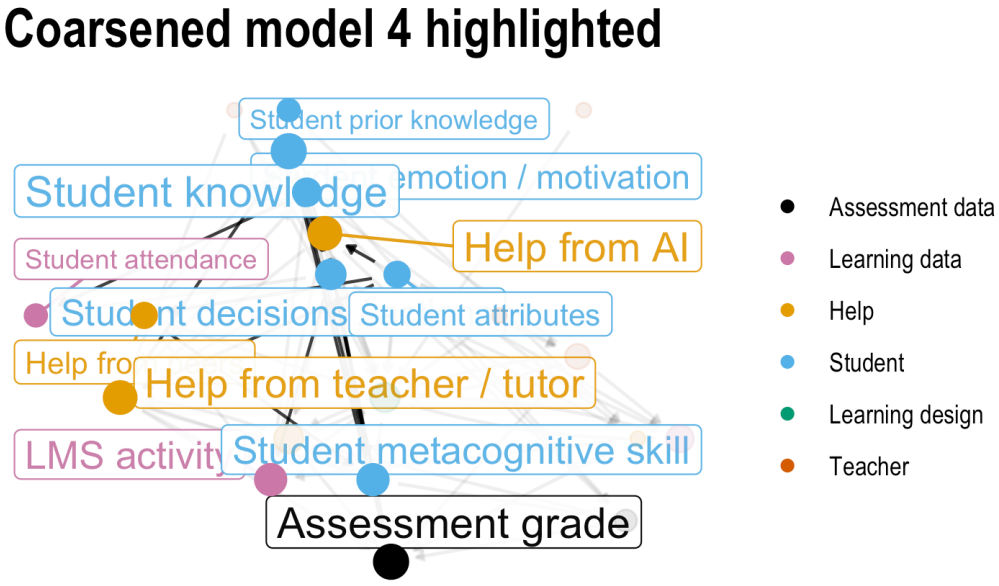

```
p_coarse_show_and_save(2)
```



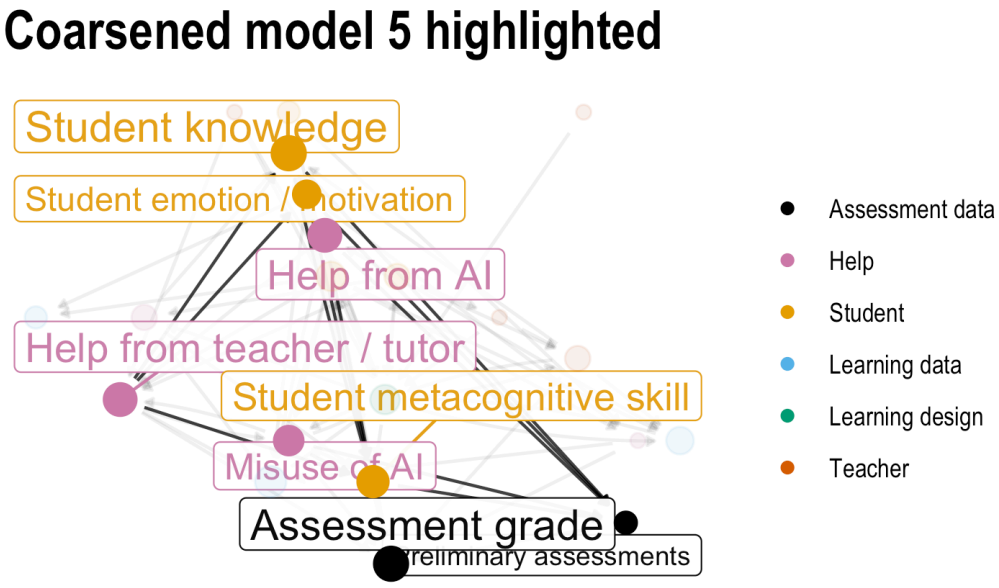

```
p_coarse_show_and_save(3)
```



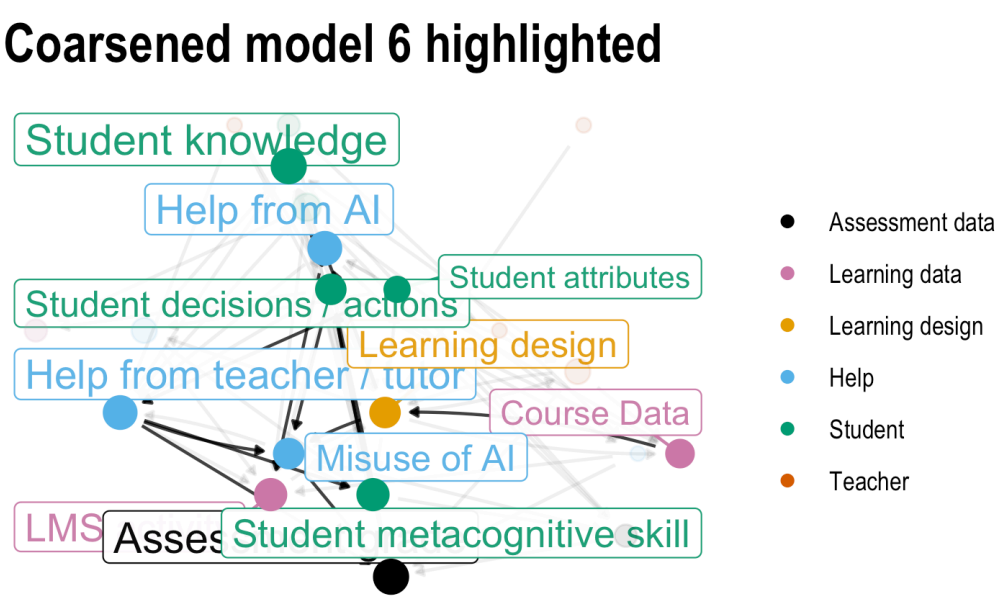

```
p_coarse_show_and_save(4)
```



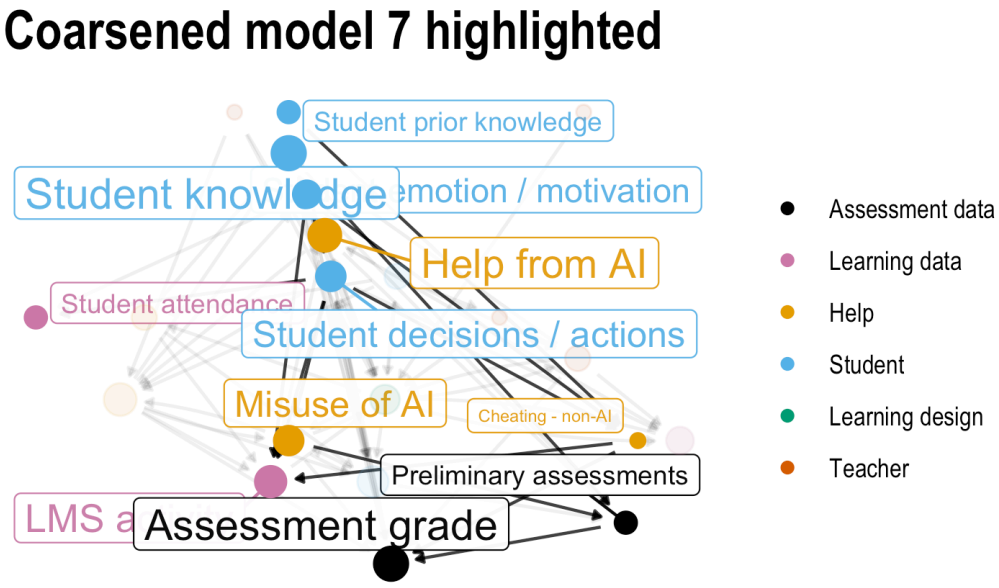

```
p_coarse_show_and_save(5)
```



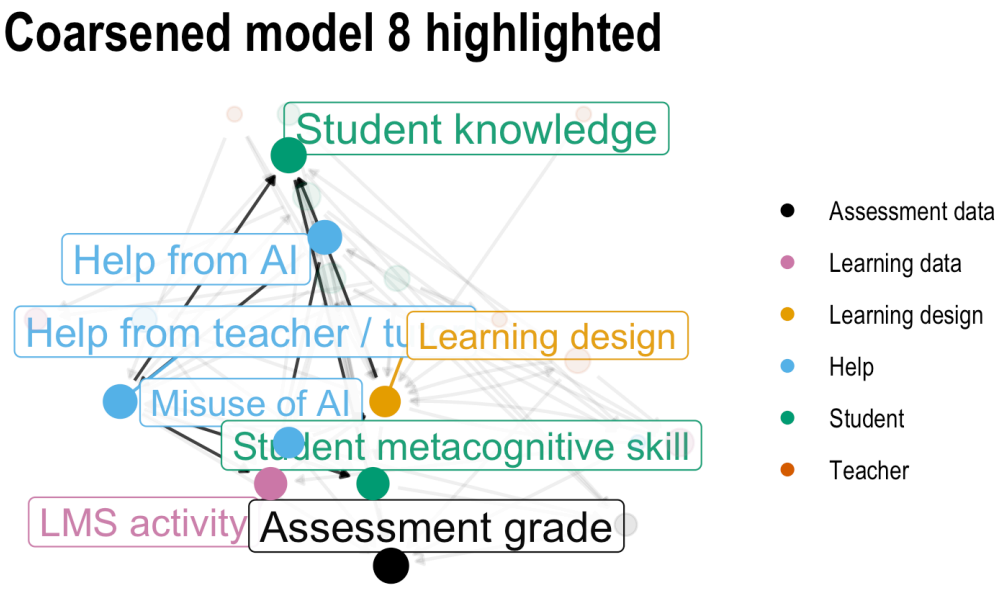

```
p_coarse_show_and_save(6)
```




```
p_coarse_show_and_save(7)
```




```
p_coarse_show_and_save(8)
```



Visualising coarsened graphs

```
merged_coarse_layout <-
    merged_all_graph |>
    g_coarsen_dag_by_ng() |>
    create_layout(layout = "sugiyama") |>
    mutate(
        x = case_when(
            name == "CD.Att" ~ 18.5,
            name == "T.Aff.WrkLd" ~ x - 5,
            name == "H.AI.good" ~ 27,
            TRUE ~ x),
        y = case_when(
            name == "CD.Att" ~ y + 1,
            name == "Grade" ~ y - 2.5,
            name == "CD.Str" ~ 3,
            name == "S.EM" ~ 5,
            TRUE ~ y)
          )
```

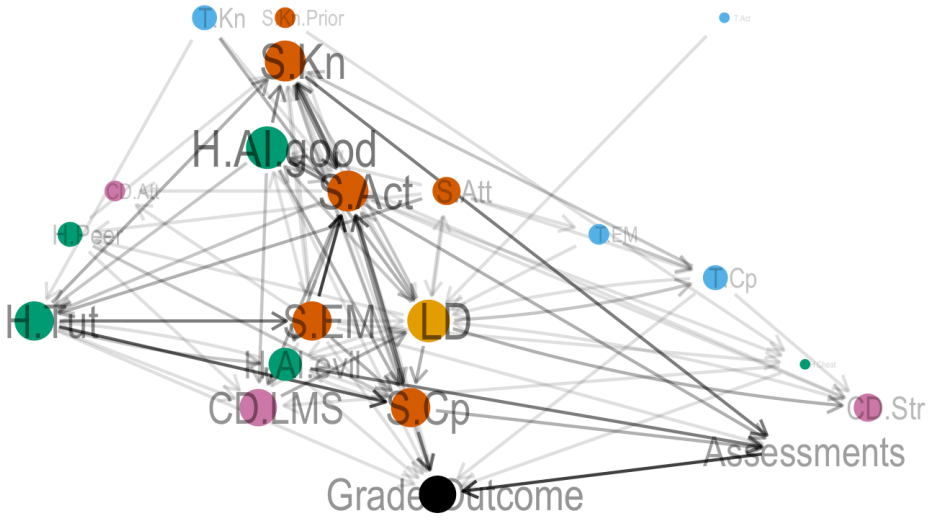
```r
p_temp <- merged_coarse_layout |>
     ggraph() +
        geom_edge_fan(aes(alpha = edge_w), strength = 0.2,
            arrow = arrow(length = unit(2, 'mm')),
            start_cap = circle(3, 'mm'),
            end_cap = circle(3, 'mm')) +
        geom_node_point(aes(colour = node_type, size = node_w)) +
    scale_color_okabe_ito(order = c(9, 7,1, 2,3, 6, 8,5, 4)) +
        # scale_color_npg(drop = FALSE) +
        # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.2,0.6)) +
        theme(legend.position = "none")

bc <- ggplot_build(p_temp)
x_limits_course <- bc$layout$panel_params[[1]]$x.range
y_limits_course <- bc$layout$panel_params[[1]]$y.range

coarse_plot_from_layout <- function(lyt) {
    lyt |>
        ggraph() +
        geom_edge_fan(aes(alpha = edge_w), strength = 0.2,
            arrow = arrow(length = unit(2, 'mm')),
            start_cap = circle(3, 'mm'),
            end_cap = circle(3, 'mm')) +
        geom_node_point(aes(colour = node_type, size = node_w)) +
        scale_color_okabe_ito(order = c(9, 7,1, 2,3, 6, 8,5, 4), drop = FALSE) +
        # ggsci::scale_color_npg(drop = FALSE) +
        # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.2,0.6)) +
        theme(legend.position = "none") +
        xlim(x_limits_course) +
        ylim(y_limits_course)
}

g_plot_coarse <- function(g, lyt = merged_coarse_layout) {
    create_layout(g, layout = "sugiyama") |>
        left_join(lyt |> select(x,y,name, node_type,node_w), by = "name",
                  suffix = c(".default", "")) |>
```

```
    coarse_plot_from_layout()
}

pCoarseMerged <- merged_coarse_layout |>
    coarse_plot_from_layout()
# ggsave("dag_merged_course.svg", pCoarseMerged, , width = 6, height = 6)
```

Note the appearance of feedback loops - these are between S.Cp and S.Kn, but also S.Cp and H.AI.evil. These metacognitive skills are clearly seen as important feedback processes in this system.

```
pCoarseMerged + geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_x = 1)
```

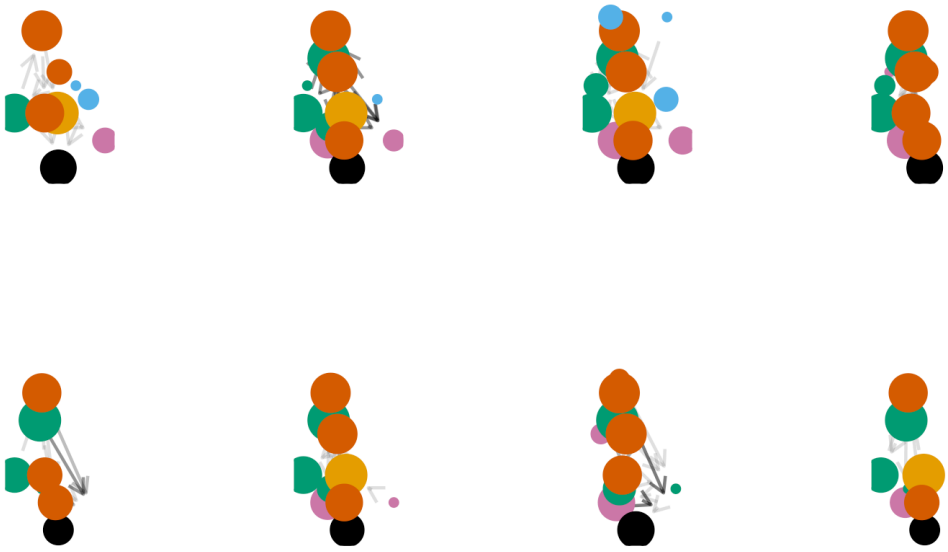
```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```




```
p1course <- g_coarsen_dag_by_ng(tdag_s1) |> g_plot_coarse()
p2course <- g_coarsen_dag_by_ng(tdag_s2) |> g_plot_coarse()
```

```
p3course <- g_coarsen_dag_by_ng(tdag_s3) |> g_plot_coarse()
p4course <- g_coarsen_dag_by_ng(tdag_s4) |> g_plot_coarse()
p5course <- g_coarsen_dag_by_ng(tdag_s5) |> g_plot_coarse()
p6course <- g_coarsen_dag_by_ng(tdag_s6) |> g_plot_coarse()
p7course <- g_coarsen_dag_by_ng(tdag_s7) |> g_plot_coarse()
p8course <- g_coarsen_dag_by_ng(tdag_s8) |> g_plot_coarse()

pAllcourse <- (p1course | p2course | p3course | p4course) / (p5course | p6course | p7course
# ggsave("dags_all_course.svg", pAllcourse, width = 18, height = 9)

pAllcourse
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



```
# all node group edges
all_edges |>
```

```
    mutate(
        from_g = code_node_group(from),
        to_g = code_node_group(to)) |>
    filter(to_g == "LD") |>
    filter(from_g == "CD.LMS")
```

```
# A tibble: 3 x 5
  Model from   to            from_g to_g
  <chr> <chr>  <chr>         <chr>  <chr>
1 s8    CD.LMS LD.Task.Code  CD.LMS LD
2 s8    CD.LMS LD.Task.Prac  CD.LMS LD
3 s8    CD.LMS LD.Task.Write CD.LMS LD
```

Figure: Consensus DAG

Most common node group patterns - nodes in at least half the models, then looking at the resulting graph of node-groups.

```
node_list_majority_of_models <-
    all_nodes |>
    count(name) |>
    filter(n >= 4) |>
    pull(name)

coarse_filtered_graph_temp <-
    merged_all_graph |>
    activate("nodes") |>
    filter(name %in% node_list_majority_of_models)

coarse_filtered_graph <- tbl_graph(
    nodes = coarse_filtered_graph_temp %N>%
        as_tibble() |>
        mutate(name = code_node_group(name),
               node_type = code_node_type(name)) ,
    edges = coarse_filtered_graph_temp %E>%
        as_tibble() %>%
        mutate(
            from = coarse_filtered_graph_temp %N>% pull(name) %>% .[from],
            to   = coarse_filtered_graph_temp %N>% pull(name) %>% .[to]
        ) %>%
        select(from, to, edge_w) |>
```

```
        mutate(across(from:to, code_node_group))) |>
    activate(edges) |>
    mutate(
        switched = map_lgl(row_number(), function(i) {
            from_i <- .E()$from[i]
            to_i   <- .E()$to[i]
            any(from == to_i & to == from_i)
        })
    )
```

```
# Old - not used
lyt_top_node_groups <- create_layout(coarse_filtered_graph, layout = "sugiyama") |>
    mutate(
        x = case_when(
            nice_name == "Student subject knowledge" ~ 0.5,
            nice_name == "Assessment grade" ~ 0.5,
            nice_name == "Help from teacher / tutor" ~ 1,
            nice_name == "Task design" ~ 0.5,
            TRUE ~ x
        ),
        y = case_when(
            nice_name == "Student subject knowledge" ~ 2.5,
            # nice_name == "Assessment grade" ~ 0.5,
            nice_name == "LMS activity" ~ 1,
            nice_name == "Task design" ~ 4,
            TRUE ~ y
        )
    )

p_top_node_groups <- ggraph(lyt_top_node_groups) +
    geom_edge_fan(aes(edge_linetype = factor(switched), alpha = edge_w, ), # alpha = edge_w,
                  strength = 0.8,
                  arrow = arrow(length = unit(2, 'mm'), type = "closed"),
                  start_cap = circle(3, 'mm'),
                  end_cap = circle(3, 'mm')) +
    geom_node_point(aes(colour = node_type, size = node_w)) +
        ggsci::scale_color_npg(drop = FALSE) +
        # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.2,0.9)) +
        theme(legend.position = "none") +
```

```
    geom_node_text(aes(label = nice_name, colour = node_type),
                   alpha = 0.9, nudge_y = -.3, repel = T)

# p_top_node_groups
# ggsave("figures/FIG--consensus-DAG.png", p_top_node_groups, dpi = 600, height = 4, width =
```

Weighting by num of models instead This one uses only nodes that are in the majority of models, then coarsens the graph, then aggregates edges based on the number of models that they appear in. The edges must also appear in at least 2 models

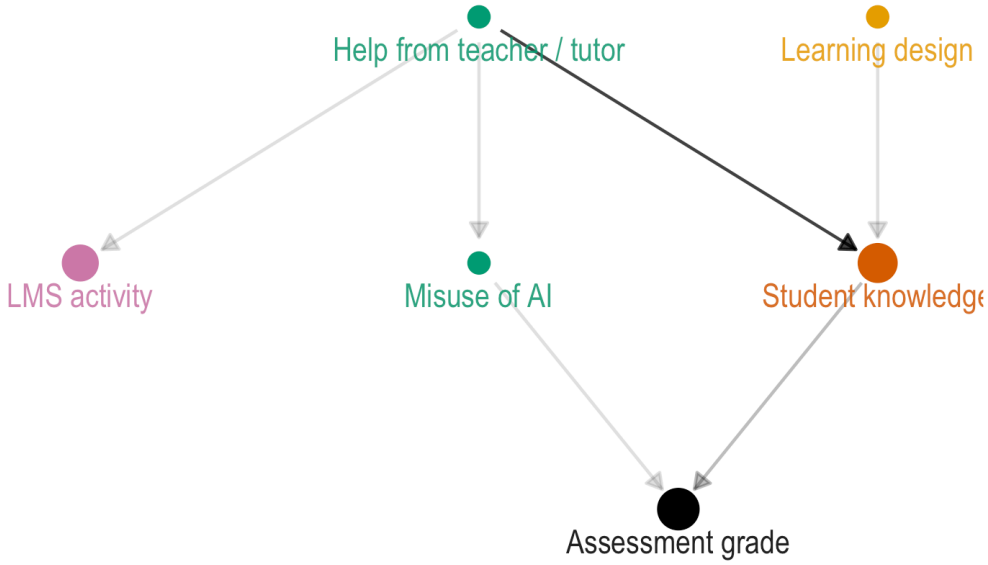
```
node_list_majority_of_models <-
    all_nodes |>
    count(name) |>
    filter(n >= 4) |>
    pull(name)

coarse_filtered_graph_ng_filled <-
    tbl_graph( # only nodes that are in most models (4 or more)
        nodes = all_nodes |>
            filter(name %in% node_list_majority_of_models) |>
            mutate( # but then coarsen into node groups
                node_group = code_node_group(name),
                   node_type = code_node_type(name)) |>
            count(node_group, node_type) |>
            rename(node_w = n),
        edges =     all_edges |> # Only edges between node groups...
            mutate(across(from:to, code_node_group)) |>
            distinct(Model, from, to) |>
            count(from, to) |>
            # filter(n > 1) |> # ...that occur in at least two models
            filter(from %in% code_node_group(node_list_majority_of_models)) |>
            filter(to %in% code_node_group(node_list_majority_of_models)) |>
            rename(edge_w = n)) |>
    activate("edges") |>
    mutate(
        switched = map_lgl(row_number(), function(i) {
            from_i <- .E()$from[i]
            to_i   <- .E()$to[i]
            any(from == to_i & to == from_i)
        })
    ) |>
```

```
    activate("nodes") |>
    inner_join(node_group_nice_name)
```

```
Joining with `by = join_by(node_group)`
```

Filtered - edges must be in at least 2 models

```
lyt_coarse_filtered_ng_filtered <- create_layout(
    coarse_filtered_graph_ng_filled |>
        activate("edges") |>
        filter(edge_w > 1) |>
        mutate(
            switched = map_lgl(row_number(), function(i) {
                from_i <- .E()$from[i]
                to_i   <- .E()$to[i]
                any(from == to_i & to == from_i)
            })
        ) ,
    layout = "sugiyama")

p_top_node_groups_filtered <- ggraph(lyt_coarse_filtered_ng_filtered) +
    geom_edge_fan(aes(edge_linetype = factor(switched), alpha = edge_w, ), # alpha = edge_w,
                  strength = 0.8,
                  arrow = arrow(length = unit(2, 'mm'), type = "closed"),
                  start_cap = circle(3, 'mm'),
                  end_cap = circle(3, 'mm')) +
    geom_node_point(aes(colour = node_type, size = node_w)) +
    scale_color_okabe_ito(order = c(9, 7,1, 2,3, 6, 8,5, 4), drop = FALSE) +
    # ggsci::scale_color_npg(drop = FALSE) +
    # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
    theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.2,0.9)) +
        theme(legend.position = "none") +
    scale_size(range = c(3,6)) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   alpha = 0.9, nudge_y = -.13, repel = F) +
    xlim(c(min(lyt_coarse_filtered_ng_filtered$x) - 0.15, max(lyt_coarse_filtered_ng_filtered

p_top_node_groups_filtered
```




```
ggsave("figures/FIG--consensus-sub-model-filtered.jpg",
       p_top_node_groups_filtered,
       dpi = 900,
       height = 4, width = 6)
```

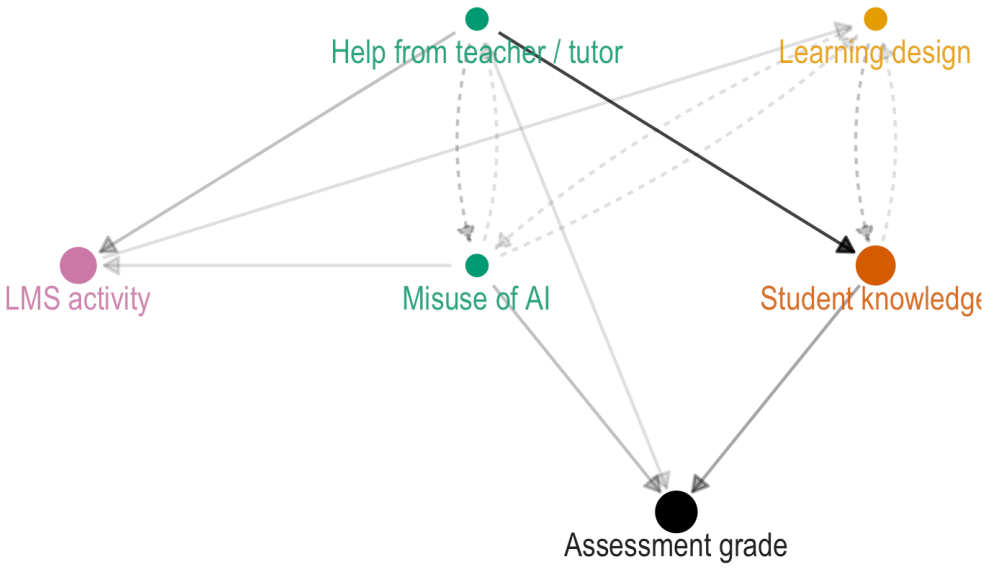
All edges between node groups from the majority nodes

```
lyt_coarse_filtered_ng_full <- create_layout(
    coarse_filtered_graph_ng_filled,
    layout = "sugiyama") |>
    select(-x, -y) |>
    inner_join(lyt_coarse_filtered_ng_filtered |> select(x, y, node_group))
```

```
Joining with `by = join_by(node_group)`
```

```
p_top_node_groups_full <- ggraph(lyt_coarse_filtered_ng_full) +
    geom_edge_fan(aes(edge_linetype = factor(switched), alpha = edge_w, ), # alpha = edge_w,
                  strength = 0.8,
                  arrow = arrow(length = unit(2, 'mm'), type = "closed"),
```

```
                  start_cap = circle(3, 'mm'),
                  end_cap = circle(3, 'mm')) +
    geom_node_point(aes(colour = node_type, size = node_w)) +
    scale_color_okabe_ito(order = c(9, 7,1, 2,3, 6, 8,5, 4), drop = FALSE) +
        # ggsci::scale_color_npg(drop = FALSE) +
        # geom_node_text(aes(label = name, size = node_w, alpha = node_w), nudge_y = -.5) +
        theme_graph() +
        scale_edge_alpha_continuous(range = c(0.15,0.8)) +
        scale_alpha_continuous(range = c(0.2,0.9)) +
        theme(legend.position = "none") +
    scale_size(range = c(3,6)) +
    geom_node_text(aes(label = nice_name, colour = node_type),
                   alpha = 0.9, nudge_y = -.13, repel = F) +
    xlim(c(
        min(lyt_coarse_filtered_ng_full$x) - 0.15,
        max(lyt_coarse_filtered_ng_full$x) + 0.15))

p_top_node_groups_full
```



```
ggsave("figures/FIG--consensus-sub-model-full.jpg",
       p_top_node_groups_full,
       dpi = 900,
       height = 4, width = 6)
```

Node groups close to the constructs: - Belonging - Mindset - Collaboration

```
# all nodes
all_nodes_with_metrics |>
    inner_join(name_node_nice_name) |>
    inner_join(node_group_nice_name |> rename(g_nice_name = nice_name)) |>
    select(name, nice_name, g_nice_name, node_group, Model, pagerank_no_outcome, pagerank, bt
    arrange(name, Model) |>
    knitr::kable()
```

```
Joining with `by = join_by(name)`
Joining with `by = join_by(node_group)`
```

name	nice_name	g_nice_name	node_group	Model	pagerank_no_outcome	pagerank	betw_rel_rank
CD	Any system data	Course Data	CD.Str	S1	0.2598017	0.1254606	5000000
CD.Att	Attendance	Student attendance	CD.Att	S4	0.0229468	0.0435203	3333333
CD.Att	Attendance	Student attendance	CD.Att	S7	0.0348776	0.0360256	6500000
CD.LMS	LMS activity	LMS activity	CD.LMS	S2	0.0489246	0.0601186	7272727
CD.LMS	LMS activity	LMS activity	CD.LMS	S3	0.1242479	0.0777304	4000000
CD.LMS	LMS activity	LMS activity	CD.LMS	S4	0.1161226	0.0715004	5333333
CD.LMS	LMS activity	LMS activity	CD.LMS	S6	0.0633664	0.0518778	8461538
CD.LMS	LMS activity	LMS activity	CD.LMS	S7	0.0403759	0.0398677	6500000
CD.LMS	LMS activity	LMS activity	CD.LMS	S8	0.0576999	0.0576577	1428571
CD.LMS.code	Code artifacts	LMS activity	CD.LMS	S7	0.0415103	0.0399127	6500000
CD.LMS.writing	Writing artifacts	LMS activity	CD.LMS	S7	0.0752913	0.0477802	6500000
CD.Str	Course Data	Course Data	CD.Str	S2	0.0693898	0.0754642	7272727
CD.Str	Course Data	Course Data	CD.Str	S3	0.0569482	0.0657501	4000000
CD.Str	Course Data	Course Data	CD.Str	S6	0.0276203	0.0375671	8461538
Grade	Assessment grade	Assessment grade	Grade.Outcome	S1	0.0000000	0.1323841	5000000
Grade	Assessment grade	Assessment grade	Grade.Outcome	S2	0.0000000	0.1642707	7272727

name	nice_name	g_nice_name	node_group	Mode	pagerank_no	pagerank	w_rel_rank
Grade	Assessment grade	Assessment grade	Grade.Outcome	S3	0.0000000	0.0732885	4000000
Grade	Assessment grade	Assessment grade	Grade.Outcome	S4	0.0000000	0.0787048	5333333
Grade	Assessment grade	Assessment grade	Grade.Outcome	S5	0.0000000	0.1730290	5294118
Grade	Assessment grade	Assessment grade	Grade.Outcome	S6	0.0000000	0.1439203	0769231
Grade	Assessment grade	Assessment grade	Grade.Outcome	S7	0.0000000	0.1275476	6500000
Grade	Assessment grade	Assessment grade	Grade.Outcome	S8	0.0000000	0.1531502	7142857
Grade.Comm	Mark - Communication	Preliminary assessments	Assessments	S7	0.0562769	0.0477904	4500000
Grade.Exam	Mark - Exam	Preliminary assessments	Assessments	S5	0.1621565	0.0893094	2941176
Grade.Interp	Mark - Interpretation	Preliminary assessments	Assessments	S7	0.1158678	0.0708185	5500000
Grade.Proc	Mark - Learning process	Preliminary assessments	Assessments	S5	0.0762812	0.0573829	3529412
Grade.Quiz	Mark - Quiz	Preliminary assessments	Assessments	S5	0.0520260	0.0453547	4117647
Grade.Refl	Mark - Learning reflection	Preliminary assessments	Assessments	S5	0.1193653	0.0888427	2352941
Grade.Tech	Mark - Technical	Preliminary assessments	Assessments	S7	0.0681343	0.0531404	4500000
H	Help (general)	Help from teacher / tutor	H.Tut	S1	0.0534232	0.0643206	3000000
H	Help (general)	Help from teacher / tutor	H.Tut	S6	0.0695564	0.0633904	3846154
H	Help (general)	Help from teacher / tutor	H.Tut	S8	0.0709553	0.0672673	0000000
H.AI	Help from AI (any)	Help from AI	H.AI.good	S3	0.0202678	0.0380242	4000000
H.AI	Help from AI (any)	Help from AI	H.AI.good	S4	0.0330791	0.0559500	2000000
H.AI.Agent	Help from AI (agent)	Help from AI	H.AI.good	S5	0.0293150	0.0325804	5294118
H.AI.Agent	Help from AI (agent)	Help from AI	H.AI.good	S8	0.0311891	0.0384384	7142857
H.AI.Fb	Help from AI (feedback)	Help from AI	H.AI.good	S5	0.0293150	0.0325804	5294118

name	nice_name	g_nice_name	node_group	Mode	pagerank_no	pagerank	w_rel_rank
H.AI.Fb	Help from AI (feedback)	Help from AI	H.AI.good	S7	0.0367908	0.0366855	1500000
H.AI.Res	Help from AI (research)	Help from AI	H.AI.good	S5	0.0293150	0.0325801	5294118
H.AI.Study	Help from AI (revision)	Help from AI	H.AI.good	S5	0.0293150	0.0325801	5294118
H.AI.Study	Help from AI (revision)	Help from AI	H.AI.good	S7	0.0367908	0.0366855	1500000
H.AI.evil	Misuse of AI	Misuse of AI	H.AI.evil	S2	0.2333173	0.0929255	6363636
H.AI.evil	Misuse of AI	Misuse of AI	H.AI.evil	S5	0.0293150	0.0325801	5294118
H.AI.evil	Misuse of AI	Misuse of AI	H.AI.evil	S6	0.1136701	0.0678082	6923077
H.AI.evil	Misuse of AI	Misuse of AI	H.AI.evil	S8	0.0311891	0.0384384	7142857
H.AI.evil.code	Misuse of AI (code)	Misuse of AI	H.AI.evil	S7	0.0367908	0.0366855	1500000
H.AI.evil.write	Misuse from AI (write)	Misuse of AI	H.AI.evil	S7	0.0367908	0.0366855	1500000
H.AI.good	Help from AI (to learn)	Help from AI	H.AI.good	S2	0.0477452	0.0595770	2727273
H.AI.good	Help from AI (to learn)	Help from AI	H.AI.good	S6	0.0841086	0.0572424	3846154
H.Ment	Help from mentor	Help from teacher / tutor	H.Tut	S5	0.0293150	0.0325801	5294118
H.Peer	Help from peers	Help from peers	H.Peer	S2	0.0477452	0.0595770	4545455
H.Peer	Help from peers	Help from peers	H.Peer	S3	0.0202678	0.0380204	4000000
H.Peer	Help from peers	Help from peers	H.Peer	S4	0.0287095	0.0489605	3333333
H.Tut	Help from teacher / tutor	Help from teacher / tutor	H.Tut	S3	0.2051369	0.0891301	1333333
H.Tut	Help from teacher / tutor	Help from teacher / tutor	H.Tut	S4	0.0287095	0.0489605	4000000
H.Tut	Help from teacher / tutor	Help from teacher / tutor	H.Tut	S5	0.0293150	0.0325801	5294118
H.Tut	Help from teacher / tutor	Help from teacher / tutor	H.Tut	S8	0.0311891	0.0384384	7142857

name	nice_name	g_nice_name	node_group	Mode	pagerank_no	pagerank_w	rel_rank
H.Tut.Fb	Feedback from teacher / tutor	Help from teacher / tutor	H.Tut	S2	0.0489246	0.0601186	3636364
H.other.evil	Cheating - non-AI	Cheating - non-AI	H.Cheat	S7	0.0458741	0.0437103	2000000
LD	Learning design	Learning design	LD	S2	0.0408079	0.0541609	7272727
LD.AIguide	Guidance on AI use	Learning design	LD	S6	0.0276203	0.0375671	8461538
LD.Task	Task design	Learning design	LD	S1	0.1516042	0.1313203	3000000
LD.Task	Task design	Learning design	LD	S3	0.0863070	0.1109038	0666667
LD.Task	Task design	Learning design	LD	S6	0.0510975	0.0563506	5384615
LD.Task	Task design	Learning design	LD	S8	0.0444445	0.0480480	2857143
LD.Task.Code	Coding task design	Learning design	LD	S8	0.0601301	0.0560501	3571429
LD.Task.Gp	Groupwork task design	Learning design	LD	S8	0.1845208	0.1225205	0714286
LD.Task.Other	Other task design	Learning design	LD	S8	0.0311891	0.0384384	7142857
LD.Task.Prac	Practical task design	Learning design	LD	S8	0.0601301	0.0560501	3571429
LD.Task.Writ	Writing task design	Learning design	LD	S8	0.0601301	0.0560501	3571429
S.Act	Student decisions / actions	Student decisions / actions	S.Act	S2	0.0477452	0.0595770	1818182
S.Act	Student decisions / actions	Student decisions / actions	S.Act	S3	0.0260103	0.0443601	5333333
S.Act	Student decisions / actions	Student decisions / actions	S.Act	S6	0.1533177	0.0965383	3000000
S.Act.AIstat	Student AI strategy	Student decisions / actions	S.Act	S7	0.0513724	0.0475500	0000000

name	nice_name	g_nice_name	node_group	Mode	pagerank_no	pagerank_w	rel_rank
S.Act.Fb	Student acts on feedback	Student decisions / actions	S.Act	S4	0.1870171	0.0908156	2666667
S.Act.Fb	Student acts on feedback	Student decisions / actions	S.Act	S6	0.1579403	0.0617006	3076923
S.Act.Fb.Pstu	Student acts on feedback between subjects	Student decisions / actions	S.Act	S6	0.0276203	0.1095307	2307692
S.Act.Frm	Student does formative tasks	Student decisions / actions	S.Act	S4	0.2311575	0.1367896	0000000
S.Act.Perf	Student performance on task	Student decisions / actions	S.Act	S3	0.2006032	0.1410574	0000000
S.Act.Time	Student commits time	Student decisions / actions	S.Act	S7	0.0423690	0.0422693	2500000
S.Aff.Integ	Student acting with integrity	Student emotion / motivation	S.EM	S1	0.0440604	0.0571705	5000000
S.Aff.Joy	Student enjoyment	Student emotion / motivation	S.EM	S5	0.0417738	0.0407302	4705882
S.Aff.Mindset	Student mindset	Student emotion / motivation	S.EM	S4	0.0178806	0.0373006	5333333
S.Aff.Mindset	Student mindset	Student emotion / motivation	S.EM	S5	0.0719866	0.0590588	0000000
S.Aff.Mindset	Student mindset	Student emotion / motivation	S.EM	S7	0.0258742	0.0307403	3500000
S.Aff.Safe	Student feels safe	Student emotion / motivation	S.EM	S5	0.0355444	0.0366572	5294118
S.Aff.Wrkld	Student workload pressure	Student emotion / motivation	S.EM	S1	0.0820852	0.0795087	3000000

name	nice_name	g_nice_name	node_group	Mode	pagerank_no	pagerank_w	rel_rank
S.Aff.WrkLd	Student workload pressure	Student emotion / motivation	S.EM	S4	0.0178806	0.0373006	5333333
S.Aff.WrkLd	Student workload pressure	Student emotion / motivation	S.EM	S7	0.0258742	0.0307403	6500000
S.Att.Eq	Student equity group	Student attributes	S.Att	S6	0.0493367	0.0516507	6153846
S.Att.HSk	Student seeks help	Student attributes	S.Att	S4	0.0254798	0.0466304	4666667
S.Att.Nro	Student neuro-divergent	Student attributes	S.Att	S4	0.0178806	0.0373006	5333333
S.Att.SocCap	Student social capital	Student attributes	S.Att	S1	0.0440604	0.0571708	5000000
S.Cp.Aca	Student academic skill	Student metacognitive skill	S.Cp	S3	0.0260103	0.0443605	2666667
S.Cp.Aca	Student academic skill	Student metacognitive skill	S.Cp	S5	0.0570710	0.0481105	1176471
S.Cp.Aca	Student academic skill	Student metacognitive skill	S.Cp	S6	0.0689204	0.0755157	7692308
S.Cp.EvJdg	Student evaluative judgement	Student metacognitive skill	S.Cp	S4	0.0178806	0.0373006	5333333
S.Cp.Fb	Student feedback literacy	Student metacognitive skill	S.Cp	S4	0.0754671	0.0819901	0666667
S.Cp.Grp	Student groupwork capability	Student metacognitive skill	S.Cp	S8	0.2169727	0.1173173	2142857
S.Cp.Ind	Student individual work capability	Student metacognitive skill	S.Cp	S8	0.0601301	0.0560501	5714286
S.Cp.Lrn	Student metacognative learning capabilities	Student metacognitive skill	S.Cp	S5	0.1277487	0.0893902	0588235
S.Cp.SRL	Student SRL	Student metacognitive skill	S.Cp	S2	0.2101735	0.1149702	0909091

name	nice_name	g_nice_name	node_group	Mode	pagerank_no	pagerank_w	rel_rank
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S1	0.0894701	0.0893356	2000000
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S2	0.1574813	0.1396450	0000000
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S3	0.1328617	0.0872409	2000000
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S4	0.1619080	0.1096473	1333333
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S5	0.0508416	0.0440395	1764706
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S6	0.1058251	0.0893201	1538462
S.Kn	Student subject knowledge	Student knowledge	S.Kn	S8	0.0601301	0.0560501	5714286
S.Kn.Coms	Student communications knowledge	Student knowledge	S.Kn	S7	0.0470086	0.0437554	5000000
S.Kn.Interp	Student interpretation knowledge	Student knowledge	S.Kn	S7	0.0936112	0.0679259	2500000
S.Kn.Prior	Student prior knowledge	Student prior knowledge	S.Kn.Prior	S4	0.0178806	0.0373046	5333333
S.Kn.Prior	Student prior knowledge	Student prior knowledge	S.Kn.Prior	S7	0.0258742	0.0307403	3500000
S.Kn.Tech	Student technical knowledge	Student knowledge	S.Kn	S7	0.0626447	0.0529267	7500000
T.Act.UseAI	Teacher use of AI	Teacher use of AI	T.Act	S3	0.0202678	0.0380242	4000000
T.Aff.Integ	Teacher acts with integrity	Teacher emotion / motivation	T.EM	S1	0.0534232	0.0643216	5000000
T.Aff.WrkLd	Teacher workload pressure	Teacher emotion / motivation	T.EM	S1	0.0440604	0.0571748	5000000
T.Cp.Inst	Teacher instruction quality	Teacher capabilities	T.Cp	S2	0.0477452	0.0595770	4545455
T.Cp.KnStu	Teacher knows students	Teacher capabilities	T.Cp	S3	0.0202678	0.0380242	4000000
T.Cp.Mark	Teacher reliably marks	Teacher capabilities	T.Cp	S1	0.1780112	0.1418202	1000000

name	nice_name	g_nice_name	node_group	Model	pagerank_no	pagerank	btw_rel_rank
T.Kn.AI	Teacher AI literacy	Teacher knowledge	T.Kn	S3	0.0202678	0.0380242	4000000
T.Kn.Ass	Teacher assessment literacy	Teacher knowledge	T.Kn	S3	0.0202678	0.0380242	4000000
T.Kn.Sub	Teacher subject knowledge	Teacher knowledge	T.Kn	S3	0.0202678	0.0380242	4000000

Node importance and measurability

```
all_node_groups_aggregated_metrics <-
    all_nodes_with_metrics |>
    group_by(Model, node_group, node_type) |>
    summarise(
        p_traced = mean(measurability == "DW"),
        measurability = (3 * mean(measurability == "DW") +
                         2 * mean(measurability == "O") +
                         mean(measurability == "PO")) / 3,
        pagerank = sum(pagerank),
              pagerank_no_outcome = sum(pagerank_no_outcome),
              btw_rel = sum(btw_rel)) |>
    group_by(node_group, node_type) |>
    mutate(
        n = n(),
        overall_pagerank = sum(pagerank / 8),
           overall_pagerank_no_outcome = sum(pagerank_no_outcome / 8),
           overall_btw = sum(btw_rel)) |>
    ungroup()
```

```
`summarise()` has grouped output by 'Model', 'node_group'. You can override
using the `.groups` argument.
```

Figure: Node importance and data availability

```
d_ng_measurability_importance <-
    all_node_groups_aggregated_metrics |>
    filter(node_group != "Grade.Outcome") |>
    inner_join(node_group_nice_name, by = "node_group") |>
```

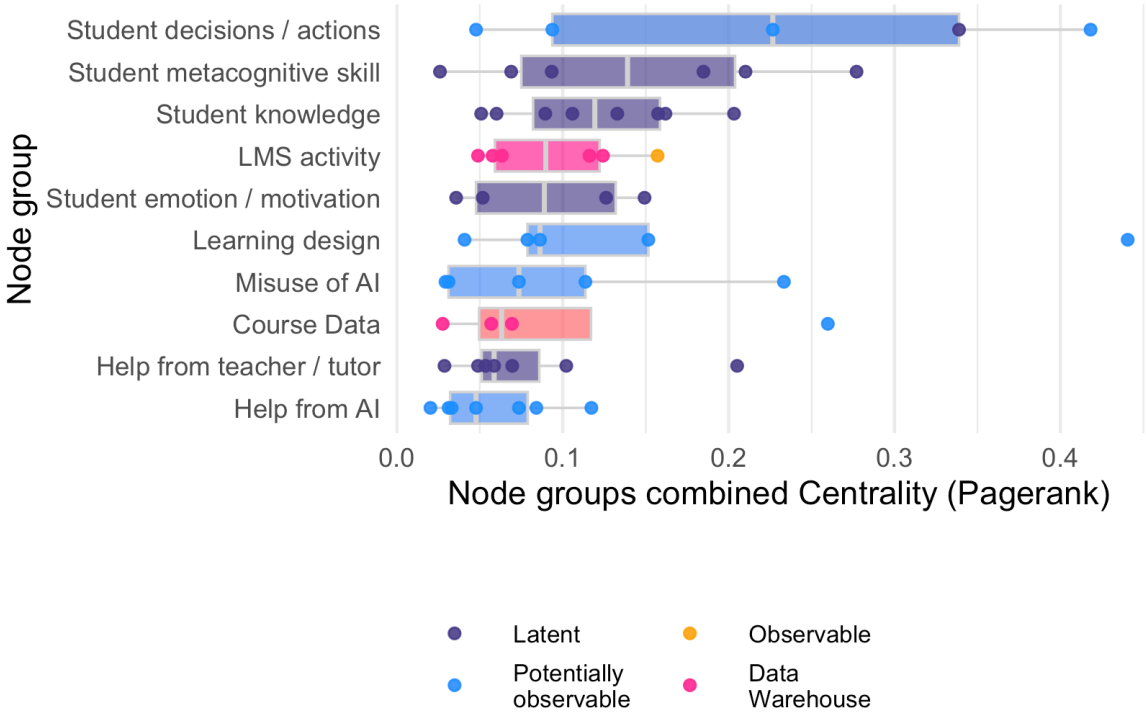
```
    mutate(node_group = nice_name) |>
    group_by(node_group) |>
    mutate(m_group = mean(measurability)) |>
    ungroup() |>
    mutate(node_group = fct_reorder(node_group, pagerank_no_outcome)) |>
    arrange(desc(node_group)) |>
    filter(n > 3) # need at least half the models with node group


p_top_node_pagerank_no_outcome_measurability <-
    d_ng_measurability_importance |>
    # arrange(Model, node_group) |>
    ggplot(aes(x = pagerank_no_outcome, y = node_group)) +
    geom_boxplot(aes(fill = m_group), colour = "lightgray", size = 0.4, alpha = 0.7) +
    # geom_line(aes(x = pagerank_no_outcome, y = node_group, group = Model), alpha = 0.6) +
    geom_point(aes(colour = measurability), alpha = 0.9) +
    scale_color_gradient(
        low = "darkslateblue",
        # mid = "darkturquoise",
        high = "aquamarine",
        # midpoint = mean(range(d_ng_measurability_importance$measurability)),
        limits = c(0,1),
        breaks = c(0, 1/3, 2/3, 1),
        labels = c("Latent","Potentially\nobservable", "Observable", "Data\nWarehouse"),
        name = "",
        guide = guide_colorbar(reverse = FALSE)
    ) +
    scale_fill_gradient(
        # midpoint = mean(range(d_ng_measurability_importance$m_group)),
        limits = c(0,1),
                low = "darkslateblue",
        # mid = "darkturquoise",
        high = "aquamarine"
        ) +
    theme_minimal() +
    ylab("Node group") +
    xlab("Node groups combined Centrality (Pagerank)") +
    # geom_rug(aes(colour = m_dist_to_trace), alpha = 0.6, sides = "b") +
    # coord_cartesian(clip = "off") +
    guides(fill = "none") +
    theme(legend.position = "bottom",
          legend.justification = "center",
```

```
    legend.box.just = "center",
          legend.text  = element_text(size = 8),
          legend.key.width = unit(1, "cm"),
          axis.ticks.y = element_blank(),
          axis.line.y = element_blank(),
          panel.grid.major.y = element_blank(),
          panel.grid.minor.y = element_blank())

# p_top_node_pagerank_no_outcome_measurability

# extract legend
leg <- get_legend(p_top_node_pagerank_no_outcome_measurability)

# arrange with relative heights (legend gets a small slice)
p_node_importance_with_measurability <- (wrap_elements(leg) /
    (p_top_node_pagerank_no_outcome_measurability + theme(legend.position = "none"))) +
  plot_layout(heights = c(0.2, 1))

# p_node_importance_with_measurability
# ggsave("figures/FIG--node_pagerank_no_outcome_measurability.png", p_node_importance_with_me
#        width = 5,
#        height = 4,
#        dpi = 900)
```

With discrete scale instead

```
m_pal <- c("darkslateblue",
    "dodgerblue","orange","deeppink")


p_top_node_pagerank_no_outcome_measurability_discrete <-
    d_ng_measurability_importance |>
    mutate(
        measurability = case_when(
            measurability < (1/6) ~ "Latent",
            measurability < 0.5 ~ "Potentially observable",
            measurability < (5/6) ~ "Observable",
            TRUE ~ "Data warehouse") |>
            factor(levels = c("Latent", "Potentially observable", "Observable", "Data wareho
    ) |>
    arrange(Model, node_group) |>
    ggplot(aes(x = pagerank_no_outcome, y = node_group)) +
```

```
    geom_boxplot(aes(fill = m_group), colour = "lightgray", size = 0.4, alpha = 0.7) +
    # geom_line(aes(x = pagerank_no_outcome, y = node_group, group = Model), alpha = 0.6) +
    geom_point(aes(colour = measurability), alpha = 0.9) +
    scale_color_manual(values = m_pal, # or retro four
                       labels = c("Latent","Potentially\nobservable", "Observable", "Data\nW
                       name = ""
    ) +
    scale_fill_gradientn(
        colours = m_pal,
        breaks = c(0, 1/3, 2/3, 5/6)
        ) +
        theme_minimal() +
            ylab("Node group") +
            xlab("Node groups combined Centrality (Pagerank)") +
            # geom_rug(aes(colour = m_dist_to_trace), alpha = 0.6, sides = "b") +
            # coord_cartesian(clip = "off") +
            guides(fill = "none",
                   color = guide_legend(nrow = 2, byrow = FALSE)) +
            theme(legend.position = "bottom",
          legend.direction = "vertical",
                  legend.justification = "left",
                  legend.text  = element_text(size = 8),
                  legend.key.width = unit(1, "cm"),
                  axis.ticks.y = element_blank(),
                  axis.line.y = element_blank(),
                  panel.grid.major.y = element_blank(),
          panel.grid.minor.y = element_blank())

p_top_node_pagerank_no_outcome_measurability_discrete
```




```
#
# # extract legend
# leg <- get_legend(p_top_node_pagerank_no_outcome_measurability_discrete)
#
# # arrange with relative heights (legend gets a small slice)
# p_node_importance_with_measurability <- (wrap_elements(leg) /
#     (p_top_node_pagerank_no_outcome_measurability_discrete + theme(legend.position = "none
#   plot_layout(heights = c(0.2, 1))


ggsave("figures/FIG--node_pagerank_no_outcome_measurability.jpg",
       p_top_node_pagerank_no_outcome_measurability_discrete,
       width = 5,
       height = 4,
       dpi = 900)
```

As table

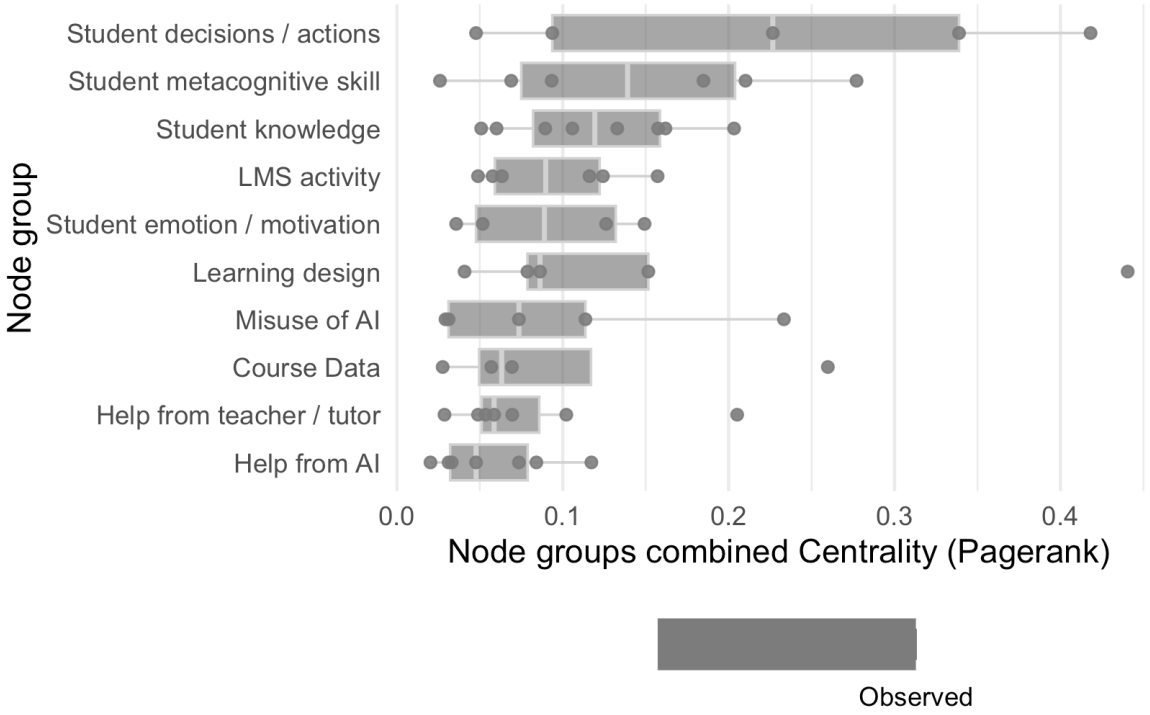
```
all_nodes_with_metrics |>
    distinct(name, measurability) |>
```

```
    janitor::tabyl(measurability) |>
    janitor::adorn_totals("row")
```

```
 measurability  n    percent
            DW  5 0.06756757
             O  9 0.12162162
            PO 26 0.35135135
             L 34 0.45945946
         Total 74 1.00000000
```

```
# Pagerank no outcome - dist to traced version - old

d_ng_trace_importance <-
    all_node_groups_aggregated_metrics |>
    inner_join(all_node_groups_distance_to_traced, by = c("Model", "node_group")) |>
    filter(node_group != "Grade.Outcome") |>
    inner_join(node_group_nice_name, by = "node_group") |>
    mutate(node_group = nice_name) |>
    group_by(node_group) |>
    mutate(p_traced_group = mean(m_dist_to_trace)) |>
    ungroup() |>
    mutate(node_group = fct_reorder(node_group, pagerank_no_outcome)) |>
    arrange(desc(node_group)) |>
    filter(n > 3) # need at least half the models with node group


p_top_node_pagerank_no_outcome_trace_dist <-
    d_ng_trace_importance |>
    # arrange(Model, node_group) |>
    ggplot(aes(x = pagerank_no_outcome, y = node_group)) +
    geom_boxplot(aes(fill = p_traced_group), colour = "lightgray", size = 0.4, alpha = 0.7) +
    # geom_line(aes(x = pagerank_no_outcome, y = node_group, group = Model), alpha = 0.6) +
    geom_point(aes(colour = m_dist_to_trace), alpha = 0.9) +
    scale_color_gradient2(
        low = "darkslateblue",
        mid = "coral",
        high = "deeppink",
        midpoint = mean(range(d_ng_trace_importance$m_dist_to_trace)),
                        breaks = range(d_ng_trace_importance$m_dist_to_trace),
                        labels = c("Observed","Latent"),
                        name = "",
```

```
        guide = guide_colorbar(reverse = TRUE)
        ) +
    scale_fill_gradient2(
        midpoint = mean(range(d_ng_trace_importance$m_dist_to_trace)),
        low = "darkslateblue",
        mid = "coral",
        high = "deeppink"
        ) +
    theme_minimal() +
    ylab("Node group") +
    xlab("Node groups combined Centrality (Pagerank)") +
    # geom_rug(aes(colour = m_dist_to_trace), alpha = 0.6, sides = "b") +
    # coord_cartesian(clip = "off") +
    guides(fill = "none") +
    theme(legend.position = "bottom",
          axis.ticks.y = element_blank(),
          axis.line.y = element_blank(),
          panel.grid.major.y = element_blank(),
          panel.grid.minor.y = element_blank())

p_top_node_pagerank_no_outcome_trace_dist
```




```
ggsave("figures/FIG--node_pagerank_no_outcome_trace_distance.jpg", p_top_node_pagerank_no_out
       width = 5,
       height = 4,
       dpi = 900)

d_ng_trace_importance |>
    group_by(node_group) |>
    summarise(m_dist_to_trace = mean(m_dist_to_trace)) |>
    arrange(desc(m_dist_to_trace))
```

```
# A tibble: 10 x 2
   node_group                    m_dist_to_trace
   <fct>                                  <dbl>
 1 Course Data                          Inf
 2 LMS activity                         Inf
 3 Student emotion / motivation           1.96
 4 Misuse of AI                           1.8
 5 Help from teacher / tutor              1.79
 6 Learning design                        1.67
 7 Student knowledge                      1.62
```

```
 8 Help from AI                          1.57
 9 Student decisions / actions           1.43
10 Student metacognitive skill           1.33
```

Above plot but lines joining models that are like?

Table: Node lists, with types, groups and importance metrics

```
# could add distance to traced
node_groups_table <-
    all_nodes_with_metrics |>
    inner_join(name_node_nice_name, by = "name") |>
    mutate(name = if_else(
        node_group == "H.AI.good" & measurability == "Traced",
        str_c(nice_name, "*"),
        nice_name
    )) |>
    inner_join(node_group_nice_name |> rename(ng_nice_name = nice_name), by = "node_group")
    mutate(node_group = ng_nice_name) |>
    group_by(node_type, node_group) |>
    mutate(measurability = case_when(
        all(measurability == "Traced") ~ "Traced",
        all(measurability == "Untraced") ~ "Untraced",
        TRUE ~ str_c("Some traced* (", sum(measurability == "Traced"), " of ", n(), ")")
    )) |>
    group_by(node_type, node_group, measurability) |>
    summarise(
        nodes = str_c(unique(name), collapse = "; "),
        n = sum(node_w),
        pagerank = round(sum(pagerank_no_outcome) / 8, 2)
        # ,btw = round(sum(btw_rel), 3)
    ) |>
    mutate(pagerank = if_else(node_group == "Grade.Outcome", NA, pagerank)) |>
    # mutate(node_type = case_when(
    #     node_type == "L" ~ "Learning Design",
    #     node_type == "S" ~ "Student",
    #     node_type == "T" ~ "Teacher",
    #     node_type == "C" ~ "Course Data",
    #     node_type == "G" ~ "Grade Data",
    #     node_type == "H" ~ "Help"
    # )) |>
    arrange(desc(n), desc(pagerank)) |>
```

```
    select(node_group, node_type, nodes, n, measurability, pagerank) |>
    ungroup()
```

```
`summarise()` has grouped output by 'node_type', 'node_group'. You can override
using the `.groups` argument.
```

```
node_groups_table
```

```
# A tibble: 21 x 6
   node_group                  node_type     nodes     n measurability pagerank
   <chr>                       <fct>         <chr> <dbl> <chr>            <dbl>
 1 Learning design             Learning des~ Task~    11 Some traced*~     0.1
 2 Help from AI                Help          Help~    11 Some traced*~     0.05
 3 Student decisions / actions Student       Stud~    10 Some traced*~     0.14
 4 Student knowledge           Student       Stud~    10 Some traced*~     0.12
 5 Student metacognitive skill Student       Stud~     9 Some traced*~     0.11
 6 Help from teacher / tutor   Help          Help~     9 Some traced*~     0.07
 7 Student emotion / motivation Student      Stud~     9 Some traced*~     0.05
 8 LMS activity                Learning data LMS ~     8 Some traced*~     0.07
 9 Assessment grade            Assessment d~ Asse~     8 Some traced*~     0
10 Preliminary assessments     Assessment d~ Mark~     7 Some traced*~     0.08
# i 11 more rows
```

```
all_nodes_with_metrics |>
    distinct(name) |>
    nrow() |>
    paste("distinct nodes")
```

```
[1] "74 distinct nodes"
```

```
all_nodes_with_metrics |>
    distinct(node_group) |>
    nrow() |>
    paste("node groups")
```

```
[1] "21 node groups"
```

```
all_nodes_with_metrics |>
    distinct(name, measurability) |> count(measurability)
```

```
# A tibble: 4 x 2
  measurability     n
  <fct>         <int>
1 DW                5
2 O                 9
3 PO               26
4 L                34
```

```
tb1 <- node_groups_table |> filter(n >= 7) |> arrange(desc(n))
tb2 <- node_groups_table |> filter(n < 7) |> arrange(desc(n))
tb1 |> clipr::write_clip()
tb2 |> clipr::write_clip()
```

What about grouping by the node category first, so it's hierarchical?

```
node_types_table <-
    node_groups_table |>
    group_by(node_type) |>
    mutate(nt = sum(n)) |>
    ungroup() |>
    arrange(desc(nt), desc(n)) |>
    select(node_type, nt, node_group, nodes, n, measurability, pagerank) |>
    mutate(node_type = str_c(node_type, " (", nt, ")")) |>
    select(-nt)

node_types_table
```

```
# A tibble: 21 x 6
   node_type    node_group                    nodes      n measurability pagerank
   <chr>        <chr>                         <chr>  <dbl> <chr>            <dbl>
 1 Student (44) Student decisions / actions   Stude~    10 Some traced*~     0.14
 2 Student (44) Student knowledge             Stude~    10 Some traced*~     0.12
 3 Student (44) Student metacognitive skill   Stude~     9 Some traced*~     0.11
 4 Student (44) Student emotion / motivation  Stude~     9 Some traced*~     0.05
 5 Student (44) Student attributes            Stude~     4 Some traced*~     0.02
 6 Student (44) Student prior knowledge       Stude~     2 Some traced*~     0.01
 7 Help (30)    Help from AI                  Help ~    11 Some traced*~     0.05
```

```
 8 Help (30)    Help from teacher / tutor    Help ~     9 Some traced*~     0.07
 9 Help (30)    Misuse of AI                 Misus~     6 Some traced*~     0.06
10 Help (30)    Help from peers              Help ~     3 Some traced*~     0.01
# i 11 more rows
```

```
tbt1 <- node_types_table |> filter(str_detect(node_type, "Student|Help"))
tbt2 <- node_types_table |> filter(!str_detect(node_type, "Student|Help"))
tbt1 |> clipr::write_clip()
tbt2 |> clipr::write_clip()
```

Edge / path importance

```
all_edge_group_with_metrics <-
    edges_with_metrics |>
    mutate(from = code_node_group(from),
           to = code_node_group(to)) |>
    group_by(from, to) |>
    summarise(
        n = sum(n),
        max_n = sum(max_n),
        edge_p_w = mean(edge_p_w),
        p_aligned = mean(switched == "aligned")
    )
```

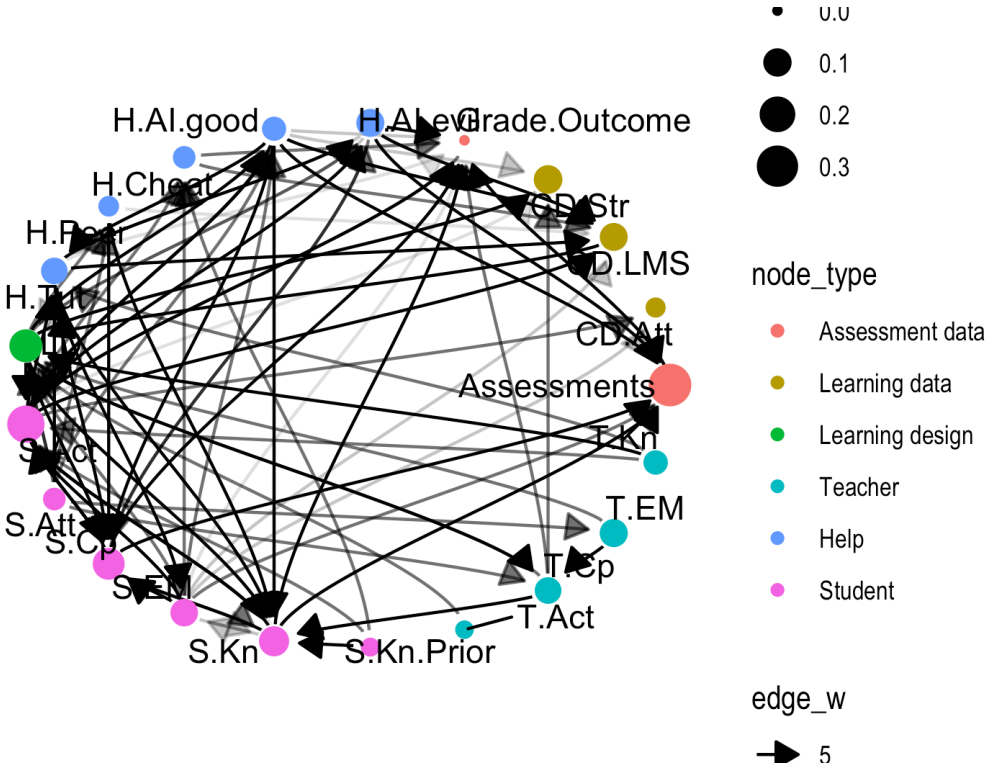
```
`summarise()` has grouped output by 'from'. You can override using the
`.groups` argument.
```

```
edge_group_analysis_edgelist <- all_edge_group_with_metrics |>
    mutate(edge_w = n + edge_p_w * n) |>
    arrange(desc(edge_w))


# the node group one here isn't one row per node group, and doesn't have the node type!
edge_group_analysis_graph <- tbl_graph(
    nodes = all_node_groups_aggregated_metrics |>
        rename(name = node_group) |>
        group_by(name, node_type) |>
        summarise(
            p_traced = mean(p_traced),
            pgrnk = mean(pagerank_no_outcome),
```

```
            node_w = sum(n)
        ) ,
    edges = edge_group_analysis_edgelist)
```

```
`summarise()` has grouped output by 'name'. You can override using the
`.groups` argument.
```

```
edge_group_analysis_graph %N>%
    arrange(name) |>
    # activate(edges) |>
    # filter(edge_w > 8) |>
    ggraph(
        layout = "circle"
        # layout = "sugiyama"
        ) +
            geom_edge_diagonal(
                aes(alpha = edge_w),
                          strength = 0.5,
                          arrow = arrow(length = unit(3, 'mm'),
                                        type = "closed"),
                          start_cap = circle(2, 'mm'),
                          end_cap = circle(3, 'mm')) +
    geom_node_point(aes(size = pgrnk, color = node_type)) +
    geom_node_text(aes(label = name), repel = TRUE) +
    scale_edge_alpha(range = c(0.1,9)) +
    theme_graph()
```



Path importance - what about the sets of implied conditional independencies? Is there a way to look at the sets of these and see if any contradict each other??